EE4530 Power System Analysis and Control

Lecture Notes on

Power Flow

2026/2027 Semester 1

Prepared by
Dr Foo Yi Shyh Eddy
Email address: eddyfoo@ntu.edu.sg
Office location: S1-B1c-89

Acknowledgement: Part of the lecture series are based on materials obtained from Dr John Chan, whose help is hereby gratefully acknowledged. Revised July 2026.

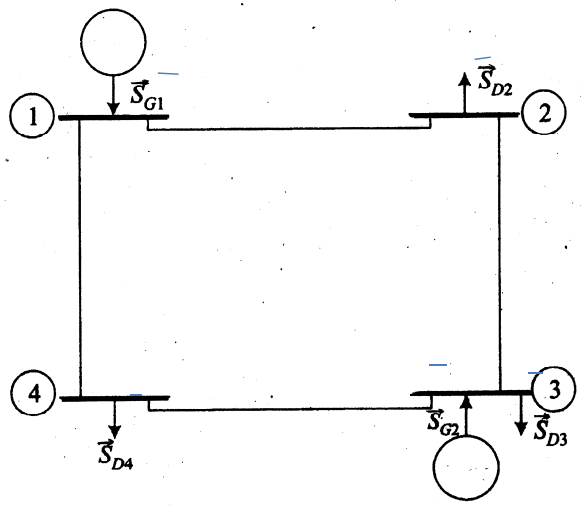
After a study on Power Flows, you will be able to:

- Understand the importance of performing power flow studies from the perspective of designing and planning power system operation in the steady-state
- Formulate power flow equations in the admittance bus frame of reference
- Solve the nonlinear algebraic power flow equations by applying some numerical methods

Power Flow Study

- Power-flow studies, commonly known as load-flow studies, are extremely important in evaluating the operations of power systems, controlling them, and planning for future expansions.
- Basically, a power-flow study yields the real and reactive power and phasor voltage at each bus on the system.
- As a consequence of a power-flow study, we can optimize the system operation with regard to system losses and load distribution.
- The effect of temporary loss of generation capacity or transmission circuits can also be investigated via a power-flow study.
- Whereas the principles of a power-flow study are straightforward, a realistic study relating to a power system can be carried out only with the digital computer.
- In such a case, numerical computations are carried out in a systematic manner by an iterative procedure.
- Two of the commonly used numerical methods are the Gauss-Seidel method and the Newton-Raphson method.
- We consider only simple examples to illustrate the principles and certain procedures of power-flow calculations.

- The power flow problem can be understood by considering a sample power system as shown.
- Buses that are supplied by generators and are called *generator buses*, e.g., buses 1 and 3.
- Other buses without generators are called *load buses,* e.g., buses 2 and 4.



$\vec{S}_{Gi}$ - complex power injected by the generator at bus i;

$\vec{S}_{Di}$ - complex power drawn by the load at bus i;

The load flow analysis will determine the complex voltage $\vec{V}_i$ at all the buses of the system, under the given operating conditions.

Power flow problem

- The power flow problem is the calculation of voltage magnitude and phase angle of the buses in the power system.
- Once these quantities are known, active and reactive power flow over lines and transformers, injection in to or drawn from buses can be computed.

Power flow equations

- The *power flow equations* are a set of equations which can be formulated from the transmission network model of the electric power system employing Kirchhoff's laws.
- The nodes of the network are the **buses**; the branches are the transmission lines that serve to interconnect the **buses**.

Types of buses

- Each bus in a power system is associated with the following four quantities:

• Magnitude of voltage, V
• Phase angle of voltage, δ
• Active power, P
• Reactive power, Q

- In a power flow problem, two out of the four quantities are specified and the remaining two are determined by the solution of power flow equations.
- Accordingly, the buses are classified into three types, namely load bus, generation or voltage-controlled bus and slack or swing or reference bus.
- The quantities specified and to be determined for the different types of buses are given in Table 1.

Table 1. Classification of buses

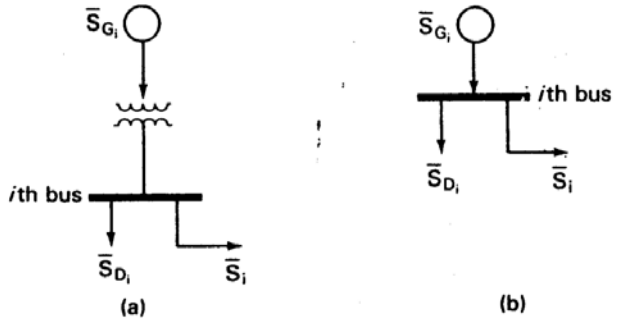
Bus	Specified quantities	Quantities to be determined
Load bus, PQ bus	P, Q	V, δ
Generation bus, PV bus	P, V	Q, δ
Slack bus, $V\delta$ bus	V, δ	P, Q

-The number of independent unknowns associated with an electric power system having one *slack bus*, n_g generator buses (which exclude the slack bus) , and a total of n system buses (all the buses) is

number of independent unknowns $= 2(n-1) - n_g$

Generator bus

- It is called a *generator bus* because it is here that a generating unit is connected.
- At each generator bus one or more generating units supply energy to local loads as well as to other loads along the transmission network.
- The situation is illustrated in Fig. 1 at the ith bus of the network.



- Fig. l(a) shows the generator connected to the bus through appropriate transformers and injecting the complex power

$$\vec{S}_{G_i} = P_{G_i} + jQ_{G_i} \qquad (1)$$

into the bus.

- $\vec{S}_{G_i}$ is then assumed to split into two parts.
- One helps to meet the needs of real and reactive power demanded by local customers and represented by

$$\vec{S}_{D_i} = P_{D_i} + jQ_{D_i} \qquad (2)$$

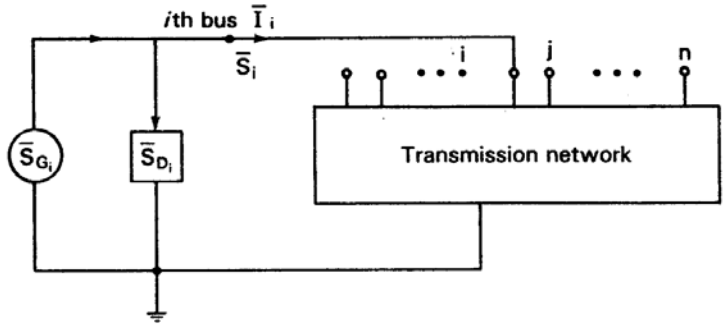
- The second part is assumed to be injected into the transmission network to help meet the power needs of other customers along the system ($\vec{S}_i$).
- This *net* outflow of complex power from the *i*th bus is represented by

$$\vec{S}_i = P_i + jQ_i = \vec{S}_{G_i} - \vec{S}_{D_i} \qquad (3)$$

- Because we are proceeding on the assumption of balanced loads throughout the system, each of the quantities in (1) to (3) can be treated on a per-phase basis for simplicity.
- Most of our calculations will be made using per unit quantities.

- Fig. (b) repeats Fig. (a) but with the transformer symbol eliminated.
- In power flow studies it is customary to lump the transformer parameters with those of the generator and to use the simpler representation.

- Finally, Fig. (c) depicts the arrangement at the *i*th bus of the electric power system and its connection to the transmission network.



(c)

- The *i*th bus is represented by a node, and the specific form of the configuration shown offers the opportunity to emphasize that as far as a power flow analysis is concerned the focus is really on finding the quantities $\vec{I}_i$ and $\vec{S}_i$ (and in turn P_i and Q_i).
- The corresponding generator quantities readily follow from eq. (3) once $\vec{S}_i$ is determined and $\vec{S}_{D_i}$, is specified.

Voltage-controlled bus

- Any bus where the *voltage magnitude can be held constant* at a specified value, is called a voltage-controlled bus.
- At the generator buses, it is possible to maintain the terminal voltage at the specified value, by adjusting the excitation of the generator.
- Also, the active power generation can be controlled by adjusting the mechanical power input.

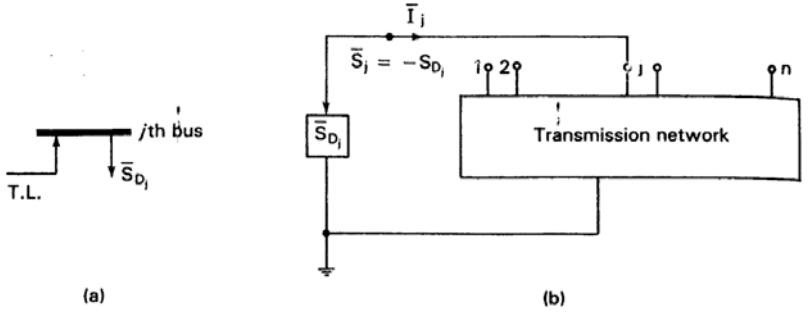
-Hence, at a generator bus, $|V|$ and P_G are specified.

-These buses are also called PV buses.

- Some load buses which have continuous reactive power capability (like having a Var compensator) are also voltage-controlled buses.

Load bus

- This next type of bus encountered in electric power systems is the *load bus.*
- There is no generating source at a load bus; instead real and reactive powers are continuously drawn from the bus to meet the needs of customers.
- The load bus is by far the most common bus to be found on the transmission network.
- The symbolic representation of the load bus is shown in Fig. (a); Fig. (b) depicts its relationship to the transmission network.



- Here it is instructive to note that the complex power *injected* into the transmission network at the *j*th load bus is

$$\vec{S}_j = -\vec{S}_{D_j} \qquad (4)$$

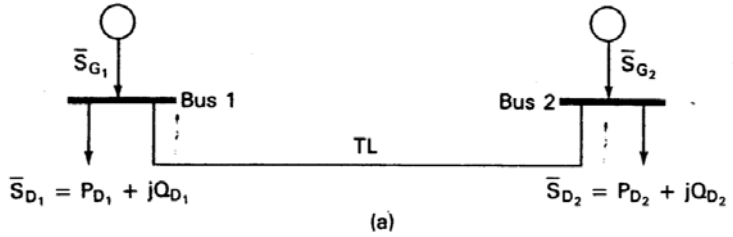
- The negative sign means that power is ***drawn from*** rather than delivered to the power system.

Slack bus

- Besides generator and load buses, the power balance condition that must prevail throughout the electric power system calls for yet another bus type.
- It is customary for the real power at each generator bus at given times of the day to be fixed for reasons of optimal economic operation.
- It is also a fact that the load demand at load buses is known.
- Rarely the sum of the demanded power exactly equalled the sum of the fixed power of the generating units.
- Consequently, rather than fix the real power generated at each generator bus, it is much more practical to do it for all except one.
- The excluded generator can then be adjusted to take up whatever slack is needed to ensure power balance.
- Because of this role the third kind of bus is called a *slack bus* and **by convention it is identified as bus 1**.

- To illustrate the application of Kirchhoff's current law in the process of developing the power flow equations, we focus first on the simple two-bus case.

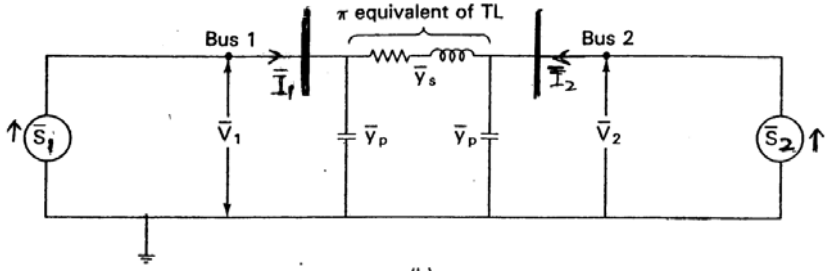
Example: *Two-bus case.* The two-bus case is depicted as shown in Fig. (a).



(a)

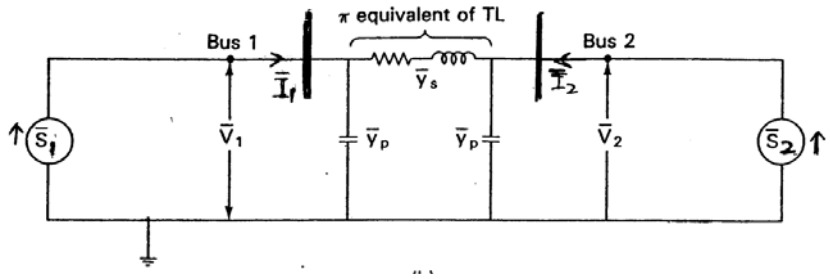
$\vec{S}_1 = \vec{S}_{G_1} - \vec{S}_{D_1} = P_1 + jQ_1$ net power injected into the bus

$\vec{S}_2 = \vec{S}_{G_2} - \vec{S}_{D_2} = P_2 + jQ_2$ net power injected into the bus



(b) all the sources are always connected to a common ground node.

- Fig. (b) is the circuit representation of the system.
- The transmission line is replaced by its π equivalent circuit and that the bus voltages to ground reference are labeled $\vec{V}_1$ and $\vec{V}_2$.
- The equivalent power source at each bus is given by $\vec{S}_1$ and $\vec{S}_2$.
- The equivalent power source at the *i*th bus *injects* current $\vec{I}_i$ into the bus.



- At the bus 1 node KCL yields

$$\vec{I}_1 = \vec{y}_p\vec{V}_1 + \vec{y}_s(\vec{V}_1 - \vec{V}_2) \qquad (5)$$

where $\vec{y}_p$ and $\vec{y}_s$ are the shunt and series admittances of the transmission line.

- Similarly, at bus 2 node,

$$\vec{I}_2 = \vec{y}_p\vec{V}_2 + \vec{y}_s(\vec{V}_2 - \vec{V}_1) \qquad (6)$$

- Rearranging gives

$$\vec{I}_1 = (\vec{y}_p + \vec{y}_s)\vec{V}_1 - \vec{y}_s\vec{V}_2 \qquad (7)$$

$$\vec{I}_2 = -\vec{y}_s\vec{V}_1 + (\vec{y}_p + \vec{y}_s)\vec{V}_2$$

- The standard form of these equations for the two-node network is

$$\vec{I}_1 = \vec{Y}_{11}\vec{V}_1 + \vec{Y}_{12}\vec{V}_2 \qquad (8)$$

$$\vec{I}_2 = \vec{Y}_{21}\vec{V}_1 + \vec{Y}_{22}\vec{V}_2$$

- where $\vec{Y}_{11}$ denotes the self-admittance as viewed from bus 1, $\vec{Y}_{22}$ is the self-admittance at bus 2, and $\vec{Y}_{12} = \vec{Y}_{21}$ is the transfer admittance between the two buses.
- Note that $\vec{Y}_{11} = \vec{y}_p + \vec{y}_s$, the sum of all admittances connected to bus 1, $\vec{Y}_{12} = \vec{Y}_{21} = -\vec{y}_s$, the negative of the sum of all admittances connected between buses 1 and 2.
- In matrix notation, (8) can written as

$$\begin{bmatrix} \vec{I}_1 \\ \vec{I}_2 \end{bmatrix} = \begin{bmatrix} \vec{Y}_{11} & \vec{Y}_{12} \\ \vec{Y}_{21} & \vec{Y}_{22} \end{bmatrix} \begin{bmatrix} \vec{V}_1 \\ \vec{V}_2 \end{bmatrix} \qquad (9)$$

or

$$\vec{I}_{bus} = \vec{Y}_{bus}\vec{V}_{bus} \qquad (10)$$

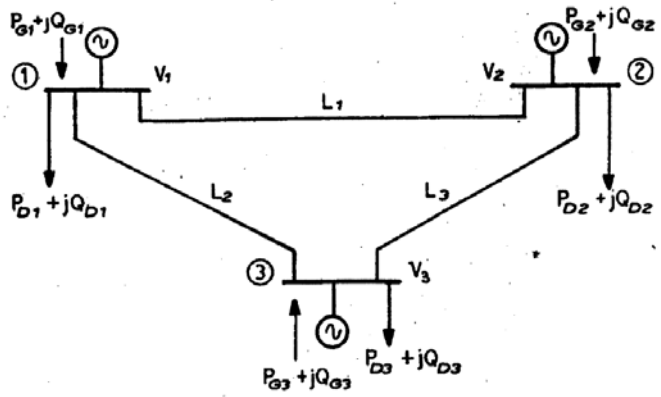
$$\vec{I}_i = \sum_{j=1}^{n} \vec{Y}_{ij}\,\vec{V}_j \qquad (11)$$

- $\vec{Y}_{bus}$ is called the ***bus admittance matrix***.

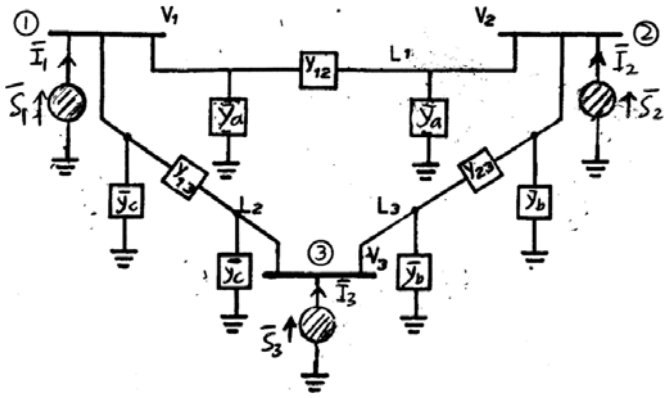
Finding the bus admittance matrix $\vec{Y}_{bus}$

- In general, the elements of the bus admittance matrix can be computed as follows:
- The diagonal element corresponding to a particular bus is the sum of all the admittances connected to that bus, i.e., $\vec{Y}_{ii} = \sum_{j=1}^{n} \vec{y}_{ij}$.
- The off-diagonal element is the negative of the admittances connected between the buses, i.e., $\vec{Y}_{ij} = \vec{Y}_{ji} = -\vec{y}_{ij},\ i \neq j$.
- $\vec{Y}_{ij}$ and $\vec{Y}_{ji}$ are 0 if there is no connection between bus i and j.

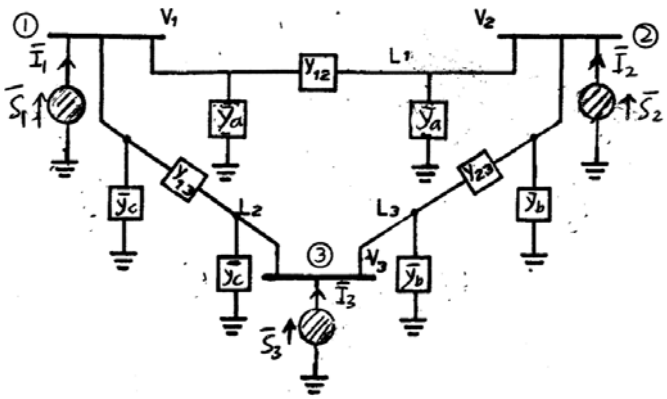
Example: *Three-bus case.* The three-bus case is depicted as shown. Obtain the bus admittance matrix $\vec{Y}_{bus}$ for the three–bus transmission network.



$$\vec{S}_1 = \vec{S}_{G_1} - \vec{S}_{D_1} = P_{G1} + jQ_{G1} - (P_{D1} + jQ_{D1})$$
$$\vec{S}_2 = \vec{S}_{G_2} - \vec{S}_{D_2} = P_{G2} + jQ_{G2} - (P_{D2} + jQ_{D2})$$
$$\vec{S}_3 = \vec{S}_{G_3} - \vec{S}_{D_3} = P_{G3} + jQ_{G3} - (P_{D3} + jQ_{D3})$$



Equivalent circuit of three-bus system



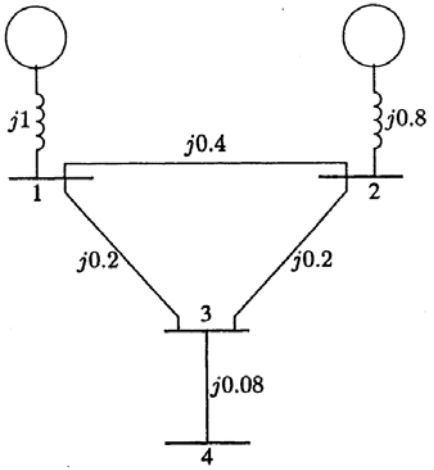
- By inspection, the diagonal elements are

$$\vec{Y}_{11} = \vec{y}_a + \vec{y}_c + \vec{y}_{12} + \vec{y}_{13}$$
$$\vec{Y}_{22} = \vec{y}_a + \vec{y}_b + \vec{y}_{12} + \vec{y}_{23}$$
$$\vec{Y}_{33} = \vec{y}_b + \vec{y}_c + \vec{y}_{13} + \vec{y}_{23}$$

- The off-diagonal elements are

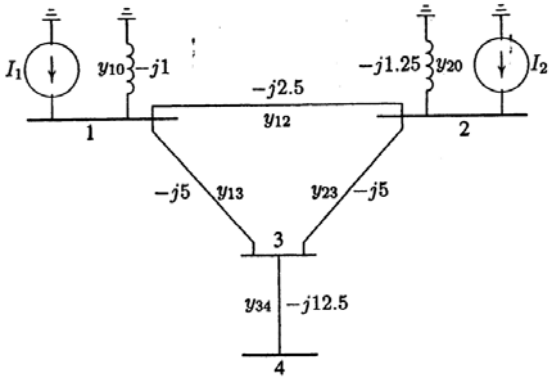
$$\vec{Y}_{12} = \vec{Y}_{21} = -\vec{y}_{12}\ ,\ \ \vec{Y}_{13} = \vec{Y}_{31} = -\vec{y}_{13},$$
$$\vec{Y}_{23} = \vec{Y}_{32} = -\vec{y}_{23}$$

Example: Consider the four-bus power system as shown where impedance are expressed in per unit on a common MVA base and for simplicity resistances are neglected. Since the nodal solution is based on Kirchhoff's current law, impedances are converted to admittance, i.e., $\vec{y}_{ij} = 1/\vec{z}_{ij}$.



Impedance diagram

- The circuit has been redrawn as shown in terms of admittances and transformation to current sources.
- Node 0 (which is normally ground) is taken as reference.



Admittance diagram

- By inspection, we get

$$\vec{Y}_{11} = \vec{y}_{10} + \vec{y}_{12} + \vec{y}_{13}$$

$$\vec{Y}_{22} = \vec{y}_{20} + \vec{y}_{12} + \vec{y}_{23}$$

$$\vec{Y}_{33} = \vec{y}_{13} + \vec{y}_{23} + \vec{y}_{34}$$

$$\vec{Y}_{44} = \vec{y}_{34}$$

$$\vec{Y}_{12} = \vec{Y}_{21} = -\vec{y}_{12}\,, \qquad \vec{Y}_{13} = \vec{Y}_{31} = -\vec{y}_{13},$$

$$\vec{Y}_{14} = \vec{Y}_{41} = 0, \qquad \vec{Y}_{24} = \vec{Y}_{42} = 0,$$

$$\vec{Y}_{23} = \vec{Y}_{32} = -\vec{y}_{23} \quad , \qquad \vec{Y}_{34} = \vec{Y}_{43} = -\vec{y}_{34}$$

- Since there is no connection between bus 1 and 4, $\vec{Y}_{14} = \vec{Y}_{41} = 0$; similarly, $\vec{Y}_{24} = \vec{Y}_{42} = 0$.

Summary of important facts

- A ***bus*** is the meeting point of various components in a power system.
- The bus is a conductor having negligible impedance.
- The bus is considered as a point of constant voltage in a power system.
- From network point of view, when one of the nodes of a power network is taken as the reference, the other nodes of the network are called *buses.*

- Each bus in a system is associated with four parameters namely magnitude and phase angle of voltage and real and reactive power.
- In a power flow problem, two out of the four quantities-magnitude and phase angle of voltage, and real and reactive power are specified and the remaining two are determined by the solution of power flow equations.
- Accordingly, the buses are classified into three types, namely load bus, generation or voltage controlled bus and slack or swing or reference bus.
- The quantities specified and to be determined for the different types of buses are given in Table 1.

- In a power system generation and load are known, *a priori,* but transmission and transformation losses will be known only after the power flow solution is complete because losses depend on the relative location and dispatch of generation with respect to load.
- It is therefore required to consider one of the generation buses as slack bus which will supply the power losses (active and reactive).

- The ***bus admittance matrix*** $\vec{Y}_{bus}$ is the matrix consisting of the self and mutual admittances of the network of a power system.
- It is formed from the nodal equations of the power system and is *symmetrical*.
- The diagonal elements of a bus admittance matrix are called *self admittances* of the buses and are obtained by adding all the admittances connected to that bus.
- The off-diagonal elements are called *mutual* or *transfer admittances* of the buses and obtained by negating the admittance between the two buses under consideration.

- Solving the power system network to obtain the voltages (magnitude and phase angle) at all the buses, power (real and reactive) flowing over all the lines and transformers, losses and power injected at all the buses is called *power flow study*.
- The voltage (magnitude and phase angle) at all the buses, power (real and reactive) flowing over all the lines and transformers, losses and power injected at all the buses is obtained from the power flow study.
- The power flow study of a power system is required to decide optimum operation of existing system with adequate security and for planning the future expansion of the system with sufficient reliability.
- It is also essential for designing a new power system.
- The solution obtained from the power flow study also gives the initial conditions of the system for carrying out fault analysis and stability studies to evaluate the transient behaviour of the system.

Derivation of power flow equations

- To derive the power flow equations, it is useful to focus on the expression for the current injected into the transmission network at the *i*th bus.
- Thus, from (11), this current is expressed as the summation $\vec{I}_i = \sum_{j=1}^{n} \vec{Y}_{ij}\,\vec{V}_j$.
- Because of the power constraints that are put on the generating units at the generator buses, it is necessary at this point to relate $\vec{I}_i$ to $\vec{S}_i$.
- We have the complex power

$$\vec{S}_i = \vec{V}_i \vec{I}_i^* \quad \text{or} \quad \vec{S}_i^* = \vec{V}_i^* \vec{I}_i \qquad (12)$$

where $\vec{S}_i$ is related to the generating power $\vec{S}_{G_i}$ through $\vec{S}_i = \vec{S}_{G_i} - \vec{S}_{D_i}$ (eqn. (3)).

- The equation for the complex conjugate power at the ith bus is (from (11) and (12))

$$\vec{S}_i^* = \vec{V}_i^* \vec{I}_i = P_i - jQ_i = \vec{V}_i^* \sum_{j=1}^{n} \vec{Y}_{ij}\,\vec{V}_j \qquad (13)$$

Expressions for P_i and Q_i

- We can get expressions for P_i and Q_i as follows.
- Let $\vec{Y}_{ij} = Y_{ij}\angle\gamma_{ij}$, where γ_{ij} is the phase angle of the net transfer admittance between the ith and jth buses.
- From the complex conjugate of $\vec{I}_i = \sum_{j=1}^{n} \vec{Y}_{ij}\,\vec{V}_j$,

$$\vec{I}_i^{*} = \sum_{j=1}^{n}(Y_{ij}\angle\gamma_{ij}\, V_j\angle\delta_j)^{*}$$

$$= \sum_{j=1}^{n} Y_{ij}V_j\angle(-\delta_j - \gamma_{ij}) \qquad (14)$$

where δ_j is the phase angle of the jth bus voltage.

- Using (14) into (12) gives the general expression for the complex power that is injected into the transmission network at the ith bus:

$$\vec{S}_i = \vec{V}_i\vec{I}_i^{*} = V_i\angle\delta_i \sum_{j=1}^{n} Y_{ij}V_j\angle(-\delta_j - \gamma_{ij}) \qquad (15)$$

or

$$\vec{S}_i = V_i \sum_{j=1}^{n} Y_{ij}V_j\angle(\delta_i - \delta_j - \gamma_{ij}) \qquad (16)$$

- Since complex quantities are involved on both sides (16), it can be rewritten to give both the real and reactive powers at the ith bus.
- Thus,

$$P_i + jQ_i = V_i\{\sum_{j=1}^{n}[\, Y_{ij}V_j\cos(\delta_i - \delta_j - \gamma_{ij}) + jY_{ij}V_j\sin(\delta_i - \delta_j - \gamma_{ij})]\,\} \qquad (17)$$

- Finally, equating the real and imaginary parts of both sides of (17), the desired expressions for the power flow equations are obtained, namely,

$$P_i = V_i \sum_{j=1}^{n} Y_{ij}V_j\cos(\delta_i - \delta_j - \gamma_{ij}) \qquad (18)$$

$$Q_i = V_i \sum_{j=1}^{n} Y_{ij}V_j\sin(\delta_i - \delta_j - \gamma_{ij}) \qquad (19)$$

- A close examination of (18) and (19) readily reveals the complexity of the power flow problem.
- These equations explicitly demonstrate that the expressions for the real and reactive powers at the ith bus are dependent upon the magnitude and phase angle of *all* the bus voltages of the transmission network.
- Moreover, although the relationship between the injected current at the ith bus and the ith bus voltage as represented by eqn. (11) is a linear one, the linearity condition does not carry over into eqns. (18) and (19).
- Consequently, the solutions of these equations cannot be found in closed form but rather are to be found via iterative procedures.

The solution process

- The immediate goal of the analysis is to find the voltage profile of the entire transmission network.
- The ensuing knowledge of the bus voltage vector essentially offers an account of the state of the electric power system.
- Because of the iterative aspect of the solution process, a step-by-step procedure is detailed.

1. Identify all the element values of the bus admittance matrix $\vec{Y}_{bus}$, based on the engineering data of the electric power system.
2. At each *load bus*, specify the pertinent values of the real and reactive powers. They are also referred to as P, Q buses.
3. At each *generator bus*, specify the pertinent values of the real power and magnitude of the bus voltage. These are also called the P, V buses.
4. At the *slack bus* (i.e., bus 1) assume $\vec{V} = 1\angle 0^o$. This assumption is in force throughout the entire solution process.
5. Next assume a voltage profile for all the remaining bus voltage phasors; i.e., assume both a magnitude and phase angle at each of these buses. A unity profile (i.e., $\vec{V}_i = 1\angle 0^o$) is assumed.

6. Introduce the information from $\vec{Y}_{bus}$ and the assumed voltage profile into (18)) to find P_i at the *i*th bus and use (19) to find Q_i. Where applicable, compare these computed values with the specified ones and determine the difference.
7. Devise a scheme whereby the difference found in step 6 can be used advantageously to correct the element values of the bus voltage vector, so that the computations of step 6 lead to significantly smaller and smaller differences between the calculated and the specified values of P and Q.
8. Repeat the iterations of the preceding step until the difference is at an acceptably low level. The corresponding voltage profile of the system then represents the solution to the power flow equations.

- In the next section, a scheme that achieves the objective of step 7 is described.

Summary of important facts

- The sequence of steps involved in power flow study is:

Step 1 Representation of the system by one-line diagram.

Step 2 Determining the impedance diagram.

Step 3 Formulation of non-linear algebraic power flow equations.

Step 4 Solution of power flow equations by an iterative method.

Guass-Siedel Method

- It is a method for solving the power flow equations.
- The inherent nonlinearity of these equations demands that the solution be achieved by a process of iteration on an appropriately modified form of the equations.

- We have the complex power (from (13))

$$\vec{S}_i^* = \vec{V}_i^* \vec{I}_i = \vec{V}_i^* \sum_{j=1}^{n} \vec{Y}_{ij} \vec{V}_j$$

- Eqn. (13) is the expression for the conjugate complex power at the *i*th bus in terms of the *n* bus voltages as well as the self-admittance at this bus and the transfer admittances to all other buses.
- Our goal here is to solve for $\vec{V}_i$.
- However because of nonlinearity and complexity, this is not possible in a closed form.
- Accordingly, it becomes necessary to resort to iteration, but first we must isolate $\vec{V}_i$ in eqn. (13).
- Thus, we rewrite the last equation as

$$\vec{S}_i^* = \vec{V}_i^* \vec{I}_i = \vec{V}_i^* (\vec{Y}_{ii} \vec{V}_i + \sum_{j=1, j\neq i}^{n} \vec{Y}_{ij} \vec{V}_j) \quad (20)$$

where the summation now no longer includes the *i*th term.

- The expression for $\vec{V}_i$ then follows as

$$\vec{V}_i = \frac{1}{\vec{Y}_{ii}}\left(\frac{\vec{S}_i^*}{\vec{V}_i^*} - \sum_{j=1, j\neq i}^{n} \vec{Y}_{ij}\,\vec{V}_j\right) \qquad (21)$$

- This last equation provides the instructions for the recursive process to be used in seeking out a solution.
- Because the iterative procedure is based on the assumption that all the quantities on the right side of the equation are known, it follows in this case that the complex conjugate of $\vec{V}_i$ is to be treated as known (or assumed) at the start of the iteration process.
- In other words, $\vec{V}_i$ is treated as the updated quantity and $\vec{V}_i^*$ as the immediate past value.
- Consequently, the recursive form of eqn. (21) becomes simply

$$\vec{V}_i^{k+1} = \frac{1}{\vec{Y}_{ii}}\left(\frac{\vec{S}_i^*}{(\vec{V}_i^*)^k} - \sum_{j=1, j\neq i}^{n} \vec{Y}_{ij}\,\vec{V}_j^k\right) \qquad (22)$$

- A straightforward application of the instructions contained on the right side of this last equation is called the *Gauss iteration method.*
- In the case of an n-bus transmission network eqn. (22) yields the following (n -1) nonlinear algebraic equations with bus 1 taken as the slack bus.

- For slack bus, its load flow equation is excluded from the calculation as $\vec{V}_1 = 1.0\angle 0^o$ (magnitude and phase known) while the 2 unknown variables P_1 and Q_1 can be calculated from (18) and (19).

$$\vec{V}_2^{k+1} = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} - (\vec{Y}_{21}\vec{V}_1^k + \vec{Y}_{23}\vec{V}_3^k + \cdots + \vec{Y}_{2n}\vec{V}_n^k)\right]$$

$$\vec{V}_3^{k+1} = \frac{1}{\vec{Y}_{33}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^k} - (\vec{Y}_{31}\vec{V}_1^k + \vec{Y}_{32}\vec{V}_2^k + \cdots + \vec{Y}_{3n}\vec{V}_n^k)\right]$$

$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \qquad (23)$$

$$\vec{V}_i^{k+1} = \frac{1}{\vec{Y}_{ii}}\left[\frac{\vec{S}_i^*}{(\vec{V}_i^*)^k} - (\vec{Y}_{i1}\vec{V}_1^k + \vec{Y}_{i2}\vec{V}_2^k + \cdots + \vec{Y}_{in}\vec{V}_n^k)\right]$$

$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$\vec{V}_n^{k+1} = \frac{1}{\vec{Y}_{nn}}\left[\frac{\vec{S}_n^*}{(\vec{V}_n^*)^k} - (\vec{Y}_{n1}\vec{V}_1^k + \vec{Y}_{n2}\vec{V}_2^k + \cdots + \vec{Y}_{n(n-1)}\vec{V}_{n-1}^k)\right]$$

- It is instructive next to focus on $\vec{V}_i^{k+1}$ in the set of eqn. (23).
- To evaluate $\vec{V}_i^{k+1}$ by Gauss's iteration procedure, the values of those bus voltages found at the kth iteration are called for.
- However, more updated information is already available for the preceding (i - 1) bus voltages.
- So it makes sense to use the most current information in the solution process.

- By this approach a more appropriate expression for $\vec{V}_i^{k+1}$ is (i = 2, 3, …n)

$$\vec{V}_i^{k+1} = \frac{1}{\vec{Y}_{ii}}\left[\frac{\vec{S}_i^*}{(\vec{V}_i^*)^k} - \sum_{j=1,}^{i-1}\vec{Y}_{ij}\,\vec{V}_j^{k+1} - \sum_{j=i+1}^{n}\vec{Y}_{ij}\,\vec{V}_j^{k}\right]$$

$$= \frac{1}{\vec{Y}_{ii}}\left[\frac{\vec{S}_i^*}{(\vec{V}_i^*)^k} - (\vec{Y}_{i1}\vec{V}_1^{k+1} + \vec{Y}_{i2}\vec{V}_2^{k+1} + \cdots\right.$$

$$\left. + \vec{Y}_{i(i-1)}\vec{V}_{i-1}^{k+1} + \vec{Y}_{i(i+1)}\vec{V}_{i+1}^{k}) + \cdots \vec{Y}_{in}\vec{V}_n^{k})\right] \qquad (24)$$

- When the iteration process is performed in accordance with the instructions contained on the right side of eqn. (24), it is called the *Gauss-Seidel iteration method.*
- The iteration process stops when $\left|\vec{V}_i^{k+1} - \vec{V}_i^{k}\right| < \epsilon$, where ϵ is the tolerance value.
- It is customary to use a value of 0.0001 pu.

Formulas to take note of

- For a *two-bus system*, with bus 1 as the slack bus,

$$\vec{V}_2^{k+1} = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1\right]$$

$$= \frac{1}{\vec{Y}_{22}}\left[\frac{P_2 - jQ_2}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1\right] \qquad (25)$$

- For a *three-bus system*, with bus 1 as the slack bus, we have

$$\vec{V}_2^{k+1} = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1 - \vec{Y}_{23}\vec{V}_3^{k}\right]$$

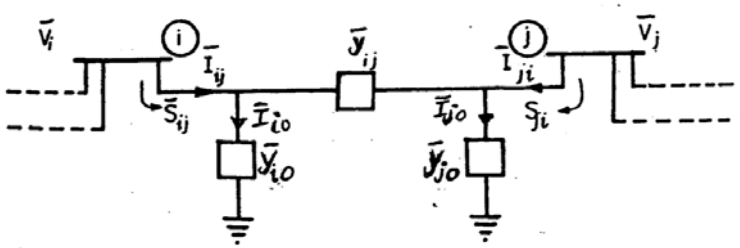
$$= \frac{1}{\vec{Y}_{22}}\left[\frac{P_2 - jQ_2}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1 - \vec{Y}_{23}\vec{V}_3^{k}\right]$$

$$\vec{V}_3^{k+1} = \frac{1}{\vec{Y}_{33}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^k} - \vec{Y}_{31}\vec{V}_1 - \vec{Y}_{32}\vec{V}_2^{k+1}\right]$$

$$= \frac{1}{\vec{Y}_{33}}\left[\frac{P_3 - jQ_3}{(\vec{V}_3^*)^k} - \vec{Y}_{31}\vec{V}_1 - \vec{Y}_{32}\vec{V}_2^{k+1}\right] \qquad (26)$$

Line flows and losses

- After the iterative solution of bus voltages, the next step is the computation of line (power) flows and line losses.
- Consider the line connecting the two buses i and j as shown.



Transmission line model for calculating line flows

- The line current $\vec{I}_{ij}$ measured at bus i and defined positive in the direction $i \rightarrow j$ is

$$\vec{I}_{ij} = \vec{I}_l + \vec{I}_{io} = \vec{y}_{ij}(\vec{V}_i - \vec{V}_j) + \vec{y}_{io}\vec{V}_i \qquad (27)$$

- Similarly, the line current $\vec{I}_{ji}$ measured at bus j and defined positive in the direction $j \rightarrow i$ is

$$\vec{I}_{ji} = -\vec{I}_l + \vec{I}_{jo} = \vec{y}_{ij}(\vec{V}_j - \vec{V}_i) + \vec{y}_{jo}\vec{V}_j \qquad (28)$$

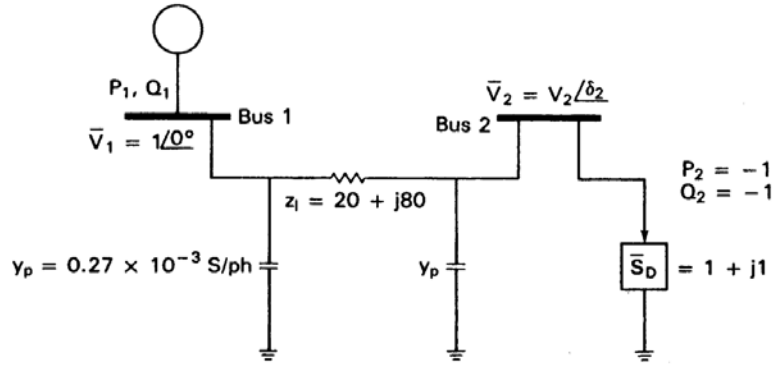
-The complex powers $\vec{S}_{ij}$ from bus i to j and $\vec{S}_{ji}$ from bus j to i are

$$\vec{S}_{ij} = \vec{V}_i\vec{I}_{ij}^{*}\,, \quad \vec{S}_{ji} = \vec{V}_j\vec{I}_{ji}^{*} \qquad (29)$$

- The power loss in line $i - j$ is the algebraic sum of the power flows determined from (29), i.e.,

$$\vec{S}_{Lij} = \vec{S}_{ij} + \vec{S}_{ji} \qquad (30)$$

Example: A two-bus system is selected to illustrate the solution procedure to find the phasor voltage at bus 2, i.e., $\vec{V}_2$. The specific two-bus system to be analyzed here for power flow along the transmission line is as shown.



Note the net powers injected at bus 1 and bus 2 are

$$\vec{S}_1 = \vec{S}_{G_1} - \vec{S}_{D_1} = \vec{S}_{G_1} - 0 = P_1 + jQ_1 \text{ pu}$$

$$\vec{S}_2 = \vec{S}_{G_2} - \vec{S}_{D_2} = 0 - \vec{S}_D = P_2 + jQ_2 = -1 - j1 \text{ pu}$$

- Because there is a single generator bus, it is made the slack bus, which means in turn that the bus 1 voltage is given by $1.0\angle 0^o$ pu.

- The second bus is a load bus at which the demanded real and reactive powers are specified.
- These quantities are identified as $P_2 = -1.0$ and $Q_2 = -1.0$ on a pu basis.
- The reference base for this power system is assumed to be 100 MVA and 230 kV at the transmission level.
- The corresponding base quantities are then

$$Z_B = \frac{(Base\ kV)^2}{Base\ MVA} = \frac{230^2}{100} = 529\ \ \Omega$$

$$Y_B = \frac{1}{Z_B} = \frac{100}{230^2} = 0.00189\ \ S$$

- The transmission line between buses 1 and 2 is 100 m long and has an impedance of

$$\vec{z}_l = 20 + j80 = 82.5\angle 76^o \ \ \Omega$$

- Express in per unit this becomes

$$\vec{z}_l = \frac{actual\ impedance}{Base\ impedance} = \frac{82.5\angle 76^o}{529} = 0.156\angle 76^o\ pu$$

- The corresponding line admittance is given as

$$\vec{y}_l = \vec{y}_{12} = \frac{1}{\vec{z}_l} = \frac{1}{0.156\angle 76^o} = 6.4105\angle -76^o$$

$$= 1.551 - j6.22$$

- The per unit value of each shunt capacitive susceptance associated with the π equivalent circuit of the transmission line is

$$\vec{y}_p = \frac{actual\ admittance}{Base\ admittance} = \frac{0.27x10^{-3}\angle 90^o}{0.00189}$$

$$= 0.1428\angle 90^o = j0.1428\ \ pu$$

- Then using $\vec{Y}_{ii} = \sum_{j=1}^{n} \vec{y}_{ij}$ and $\vec{Y}_{ij} = \vec{Y}_{ji} = -\vec{y}_{ij}$

$$\vec{Y}_{11} = \vec{Y}_{22} = \vec{y}_{12} + \vec{y}_p = Y_{11}\angle\gamma_{11} = Y_{22}\angle\gamma_{22}$$

$$= 1.551 - j6.22 + j0.1428$$

$$= 1.551 - j6.0772 = 6.272\angle -75.683^o$$

$$\vec{Y}_{12} = \vec{Y}_{21} = -\vec{y}_{12} = Y_{12}\angle\gamma_{12} = Y_{21}\angle\gamma_{21}$$

$$= -1.551 + j6.22 = 6.4105\angle 104^o$$

- These quantities completely specify the bus admittance matrix

$$\vec{Y}_{bus} = \begin{bmatrix} \vec{Y}_{11} & \vec{Y}_{12} \\ \vec{Y}_{21} & \vec{Y}_{22} \end{bmatrix} = \begin{bmatrix} Y_{11}\angle\gamma_{11} & Y_{12}\angle\gamma_{12} \\ Y_{21}\angle\gamma_{21} & Y_{22}\angle\gamma_{22} \end{bmatrix}$$

- Of course, for more buses, the preliminary work in collecting the data in preparation for a power flow study is many times greater, but the process is the same.

- The bus admittance matrix for the two-bus transmission network is

$$\vec{Y}_{bus} = \begin{bmatrix} 6.272\angle -75.683^o & 6.4105\angle 104^o \\ 6.4105\angle 104^o & 6.272\angle -75.683^o \end{bmatrix}$$

- The injected complex power at bus 2 is specified as $\vec{S}_2 = -1 - j$ (power drawn).
- Hence, $\vec{S}_2^* = -1 + j = \sqrt{2}\angle 135^o$.
- Of course, $\vec{V}_1 = 1\angle 0^o$.
- From (25), we have the recursive formula:

$$\vec{V}_2^{k+1} = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1\right]$$

$$= \frac{\vec{S}_2^*/\vec{Y}_{22}}{(\vec{V}_2^*)^k} - \frac{\vec{Y}_{21}}{\vec{Y}_{22}}1\angle 0^o\ ] \qquad (31)$$

- Using the parameter values obtained earlier gives

$$\vec{V}_2^{k+1} = \frac{\sqrt{2}\angle 135^o}{6.272\angle -75.683^o(\vec{V}_2^*)^k} - \frac{6.4105\angle 104^o}{6.272\angle -75.683^o}1\angle 0^o$$

$$= \frac{0.2255\angle 210.683^o}{(\vec{V}_2^*)^k} - 1.0221\angle 179.683^o$$

$$= \frac{0.2255\angle 210.683^o}{(\vec{V}_2^*)^k} + 1.0221 - j0.0056 \qquad (32)$$

- Begin the iteration by assuming the initial value for bus 2 voltage of unity, i.e., $(\vec{V}_2^*)^0 = 1\angle 0^o$.

1st iteration ($k = 0$):

$$\vec{V}_2^1 = \frac{0.2255\angle 210.683^o}{1\angle 0^o} + 1.0221 - j0.0056$$

$$= -0.1939 - j0.1151 + 1.0221 - j0.0056$$

$$= 0.8281 - j0.1207 = 0.8369\angle -8.293^o$$

2nd iteration ($k = 1$): $(\vec{V}_2^*)^1 = 0.8369\angle 8.293^o$
(change in sign due to the complex conjugate)

$$\vec{V}_2^2 = \frac{0.2255\angle 210.683^o}{0.8369\angle 8.293^o} + 1.0221 - j0.0056$$

$$= 0.781\angle -7.974^o$$

- Using (32) in succession gives the table below.

Iteration	$\vec{V}_2 = 1\angle 0^o$ pu
0	1.0 + j0
1	0.8281 - j0.1207
2	0.7729 - j0.1083
3	0.7555 - j0.1172
4	0.7484 - j0.1155
5	0.7458 - j0.1167
6	0.7447 - j0.1165
7	0.7443 - j0.1167
8	0.7441 - j0.1166
9	0.7441 - j0.1167
10	0.7440 - j0.1167
11	0.7440 - j0.1167

- Notice that convergence is achieved in 11 iterations (quite slow).
- For other data such as a greater load, it may take more iterations to converge to the solution.
- Of course, in some cases convergence may not be achieved because a solution may not exist or the starting point of the iteration process may not be appropriate.

- In summary, then, the solution to the power flow problem as formulated by (25) is given by the bus 2 voltage vector,

$$\vec{V}_2 = V_2 \angle \delta_2 = 0.753 \angle -8.914^o \quad \text{pu}$$

- The information can now be used to determine the real and reactive power to be supplied by the generator at the slack bus in order to meet the complex power needs along this two-bus transmission network.
- The power at the *i*th bus can be found from (13) as

$$\vec{S}_i^* = \vec{V}_i^* \vec{I}_i = P_i - jQ_i = \vec{V}_i^* \sum_{j=1}^{n} \vec{Y}_{ij} \vec{V}_j$$

- The slack bus power is

$$\vec{S}_1^* = P_1 - jQ_1 = \vec{V}_1^* \sum_{j=1}^{2} \vec{Y}_{1j} \vec{V}_j$$

$$= \vec{V}_1^* [\vec{Y}_{11}\vec{V}_1 + \vec{Y}_{12}\vec{V}_2]$$

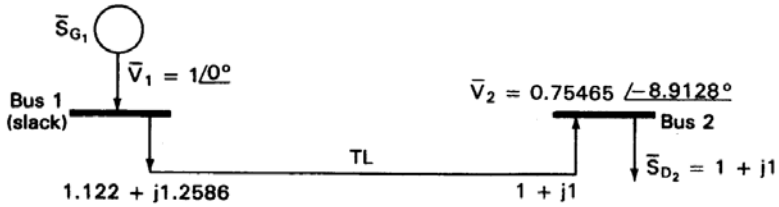
$$= 1[\, 6.272 \angle -75.683^o (1) +$$

$$(6.4105 \angle 104^o)(0.753 \angle -8.914^o)] \,]$$

$$= 1.1221 - j1.269 \quad \text{pu}$$

- Thus, $P_1 = 1.122$ pu and $Q_1 = 1.269$ pu.

-The solution to this two-bus power flow problem is summarized in the figure as shown.



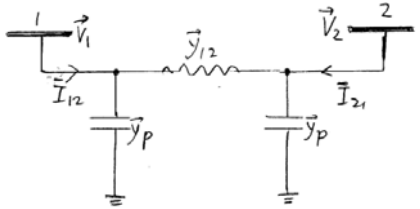
Solution to a two-bus power flow problem

- The generator at the slack bus injects a complex power of

$$\vec{S}_{G_1} = P_1 + jQ_1 = 1.122 + j1.269 \quad \text{pu}$$

into the transmission line, which according to the engineering data has line resistance, line inductance, and line capacitance.

Line loss



- Now from the figure,

$$\vec{I}_{12} = \vec{y}_p\vec{V}_1 + \vec{y}_{12}(\vec{V}_1 - \vec{V}_2)$$

$$= (j0.1428)(1) +$$

$$(6.4105\angle -76^o)(1 - 0.753\angle -8.914^o)$$

$$= 1.123 - j1.269 = 1.695\angle -48.49^o$$

$$\vec{I}_{21} = \vec{y}_p\vec{V}_2 - \vec{y}_{12}(\vec{V}_1 - \vec{V}_2)$$

$$= (j0.1428)(0.753\angle -8.914^o)$$

$$-(6.4105\angle -76^o)(1 - 0.753\angle -8.914^o)$$

$$= -1.106 + j1.518 = 1.878\angle 126.07^o$$

$$\vec{S}_{12} = \vec{V}_1\vec{I}_{12}^* = 1(1.695\angle -48.49^o)^*$$

$$= 1.695\angle 48.5^o = 1.123 + j1.269$$

$$\vec{S}_{21} = \vec{V}_2\vec{I}_{21}^*$$

$$= (0.753\angle -8.914^o)(1.878\angle 126.07^o)^*$$

$$= 1.414\angle -135.0^o = -1.0 - j1.0$$

- Thus, the power loss in the line is given by

$$\vec{S}_{L12} = \vec{S}_{12} + \vec{S}_{21}$$

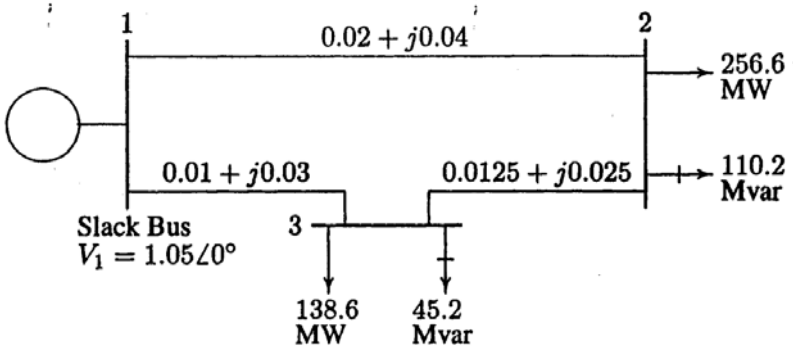
$$= 1.12264 + j1.269 - 1.0 - j1.0$$

$$= 0.123 + j0.269 = P_l + jQ_l$$

Note that in this simple case, $\vec{S}_{G_1} = \vec{S}_{L12} + \vec{S}_{D_2}$.

- $P_l = 0.123$ represents the power that is consumed as a copper loss along the transmission line.
- Similarly, the reactive power of the transmission line is $Q_l = 0.269$.

Example: The figure shows the one-line diagram of a simple three-bus power system with generator at bus 1. The magnitude of voltage at bus 1 is adjusted to 1.05 pu. The scheduled loads are buses 2 and 3 are as marked on the diagram. Line impedances are marked in pu on a 100 MVA and the line charging susceptances (shunt elements) are neglected.

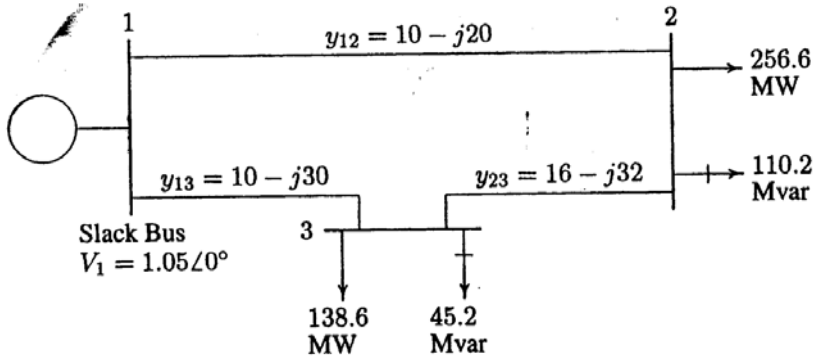


(a) Using the Gauss-Seidel method, determine the phasor values of the voltage at the load buses 2 and 3 (P-Q buses).
(b) Find *the* slack bus real and reactive power.
(c) Determine the line flows and line losses.

(a) Line impedances are converted to admittances

$$\vec{y}_{12} = \frac{1}{0.02+j0.04} = 10 - j20$$

- Similarly, $\vec{y}_{13} = 10 - j30$ and $\vec{y}_{23} = 16 - j32$.
- The admittances are marked on the network as shown.



- The complex loads in pu at buses 2 and 3 are

$$\vec{S}_2 = -\frac{Actual\ MVA}{Base\ MVA} = -\frac{256.6+j110.2}{100}$$

$$= -2.566 - j1.102\ \ pu \quad \text{(power drawn)}$$

$$\vec{S}_3 = -\frac{Actual\ MVA}{Base\ MVA} = -\frac{138.6+j45.2}{100}$$

$$= -1.386 - j0.452\ \ pu \quad \text{(power drawn)}$$

- Bus 1 is taken as the slack bus, then use

$$\vec{V}_2^0 = 1\angle 0^o, \quad \vec{V}_3^0 = 1\angle 0^o$$

- Then using (25) for a three-bus power system,

$$\vec{V}_2^{k+1} = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1 - \vec{Y}_{23}\vec{V}_3^k\right]$$

$$\vec{V}_3^{k+1} = \frac{1}{\vec{Y}_{33}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^k} - \vec{Y}_{31}\vec{V}_1 - \vec{Y}_{32}\vec{V}_2^{k+1}\right]$$

- $\vec{V}_2$ and $\vec{V}_3$ are computed as follows:

$$\vec{V}_2^{k+1} = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} - \vec{Y}_{21}\vec{V}_1 - \vec{Y}_{23}\vec{V}_3^k\right]$$

$$= \frac{1}{\vec{y}_{12}+\vec{y}_{23}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^k} + \vec{y}_{12}\vec{V}_1 + \vec{y}_{23}\vec{V}_3^k\right]$$

$$\vec{V}_3^{k+1} = \frac{1}{\vec{Y}_{33}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^k} - \vec{Y}_{31}\vec{V}_1 - \vec{Y}_{32}\vec{V}_2^{k+1}\right]$$

$$= \frac{1}{\vec{y}_{13}+\vec{y}_{23}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^k} + \vec{y}_{13}\vec{V}_1 + \vec{y}_{23}\vec{V}_2^{k+1}\right]$$

- Using the data obtained, for the first iteration, we have

$$\vec{V}_2^1 = \frac{1}{\vec{y}_{12}+\vec{y}_{23}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^0} + \vec{y}_{12}\vec{V}_1 + \vec{y}_{23}\vec{V}_3^0\right]$$

$$= \frac{1}{26-j52}\left[\frac{-2.566+j1.102}{1} + (10-j20)(1.05)\right.$$

$$\left. + (16-j32)(1)\right] = 0.9825 - j0.0310$$

$$\vec{V}_3^1 = \frac{1}{\vec{y}_{13}+\vec{y}_{23}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^0} + \vec{y}_{13}\vec{V}_1 + \vec{y}_{23}\vec{V}_2^1\right]$$

$$= \frac{1}{26-j62}\left[\frac{-1.386+j0.452}{1} + (10-j30)(1.05)\right.$$

$$\left. + (16-j32)(0.9825-j0.0310)\right]$$

$$= 1.0011 - j0.0353$$

- For the second iteration, we have

$$\vec{V}_2^2 = \frac{1}{\vec{y}_{12}+\vec{y}_{23}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^1} + \vec{y}_{12}\vec{V}_1 + \vec{y}_{23}\vec{V}_3^1\right]$$

$$= \frac{1}{26-j52}\left[\frac{-2.566+j1.102}{0.9825+j0.0310} + (10-j20)(1.05)\right.$$

$$\left. + (16-j32)(1.0011-j0.0353)\right]$$

$$= 0.9816 - j0.052$$

$$\vec{V}_3^2 = \frac{1}{\vec{y}_{13}+\vec{y}_{23}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^1} + \vec{y}_{13}\vec{V}_1 + \vec{y}_{23}\vec{V}_2^2\right]$$

$$= \frac{1}{26-j62}\left[\frac{-1.386+j0.452}{1.0011+j0.0353} + (10-j30)(1.05)\right.$$

$$\left. + (16-j32)(0.9816-j0.052)\right]$$

$$= 1.0008 - j0.0459$$

- The process is continued and a solution is converged in seven iterations as given in the table below.

Iteration	$\vec{V}_2 = 1\angle 0^o$ pu	$\vec{V}_3 = 1\angle 0^o$ pu
0	1.0 + j0	1.0 + j0
1	0.9825 - j0.0310	1.0011 - j0.0353
2	0.9816 - j0.0521	1.0008 - j0.0460
3	0.9808 - j0.0578	1.0004 - j0.0489
4	0.9803 - j0.0594	1.0002 - j0.0497
5	0.9802 - j0.0598	1.0001 - j0.0499
6	0.9801 - j0.0599	1.0001 - j0.0499
7	0.9801 - j0.0599	1.0001 - j0.0499

- The final solution is

$$\vec{V}_2 = 0.9801 - j0.06 \approx 0.9818\angle -3.50^o$$

$$\vec{V}_3 = 1.0001 - j0.05 \approx 1\angle -2.86^o$$

(b) The complex power at the *i*th bus is obtained from (13) as

$$\vec{S}_i^* = \vec{V}_i^* \vec{I}_i = P_i - jQ_i = \vec{V}_i^* \sum_{j=1}^{n} \vec{Y}_{ij}\, \vec{V}_j$$

- With the knowledge of all bus voltages, the slack bus power is obtained given by

$$P_1 - jQ_1 = \vec{V}_1^* [\vec{Y}_{11}\vec{V}_1 + \vec{Y}_{12}\vec{V}_2 + \vec{Y}_{13}\vec{V}_3]$$

$$= 1.05[(\vec{y}_{12} + \vec{y}_{13})\vec{V}_1 - \vec{y}_{12}\vec{V}_2 - \vec{y}_{13}\vec{V}_3]$$

$$= 1.05[\,(20 - j50)(1.05) -$$

$$(10 - j20)(0.98 - j0.6) -$$

$$(10 - j30)(1.0 - j0.05)]$$

$$= 4.095 - j1.89 \quad \text{pu}$$

- Thus, the slack bus real and reactive powers are $P_1 = 4.095\ pu = 409.5\ MW$ (base MVA = 100) and $Q_1 = 1.89\ \ pu = 189\ Mvar$.

(c) To find the line flows, first the line currents are computed.

- With the shunt elements neglected, the line currents in pu are

$$\vec{I}_{12} = \vec{y}_{12}(\vec{V}_1 - \vec{V}_2) = -\vec{I}_{21}$$

$$= (10 - j20)[1.05 - (0.98 - j0.06)]$$

$$= 1.9 - j0.8 \qquad \Rightarrow \vec{I}_{21} = -1.9 + j0.8$$

$$\vec{I}_{13} = \vec{y}_{13}(\vec{V}_1 - \vec{V}_3) = -\vec{I}_{31}$$

$$= (10 - j30)[1.05 - (1.0 - j0.05)]$$

$$= 2.0 - j1.0 \qquad \Rightarrow \vec{I}_{31} = -2.0 + j1.0$$

$$\vec{I}_{23} = \vec{y}_{23}(\vec{V}_2 - \vec{V}_3) = -\vec{I}_{32}$$

$$= (16 - j32)[(0.98 - j0.06) - (1.0 - j0.05)]$$

$$= -0.64 + j0.48 \qquad \Rightarrow \vec{I}_{32} = 0.64 - j0.48$$

- The line flows in pu are

$$\vec{S}_{12} = \vec{V}_1 \vec{I}_{12}^{*} = (1.05)(1.9 + j0.8)$$

$$= 1.995 + j0.84 = 199.5 \text{ MW} + \text{j}84.0 \text{ Mvar}$$

$$\vec{S}_{21} = \vec{V}_2 \vec{I}_{21}^{*} = (0.98 - j0.06)(-1.9 - j0.8)$$

$$= -1.91 - j0.67 = -191.0 \text{ MW} - \text{j}67.0 \text{ Mvar}$$

$$\vec{S}_{13} = \vec{V}_1 \vec{I}_{13}^{*} = (1.05)(2.0 + j1.0)$$

$$= 2.1 + j1.05 = 210.0 \text{ MW} + \text{j}105.0 \text{ Mvar}$$

$$\vec{S}_{31} = \vec{V}_3 \vec{I}_{31}^{*} = (1.0 - j0.05)(-2.0 - j1.0)$$

$$= -2.05 - j0.9 = -205.0 \text{ MW} - \text{j}90.0 \text{ Mvar}$$

$$\vec{S}_{23} = \vec{V}_2 \vec{I}_{23}^{*} = (0.98 - j0.06)(-0.656 + j4.8)$$

$$= -0.656 - j0.432 = -65.6 \text{ MW} - \text{j}43.2 \text{ Mvar}$$

$$\vec{S}_{32} = \vec{V}_3 \vec{I}_{32}^{*} = (1.0 - j0.05)(0.64 + j4.8)$$

$$= 0.664 + j0.448 = 66.4 \text{ MW} + \text{j}44.8 \text{ Mvar}$$

- The line losses in pu are

$$\vec{S}_{L12} = \vec{S}_{12} + \vec{S}_{21} = 1.995 + j0.84 - 1.91 - j0.67)$$

$$= 0.085 + j0.17 = 8.5\ MW + j17.0\ Mvar$$

$$\vec{S}_{L13} = \vec{S}_{13} + \vec{S}_{31} = 2.1 + j1.05 - 2.05 - j0.9$$

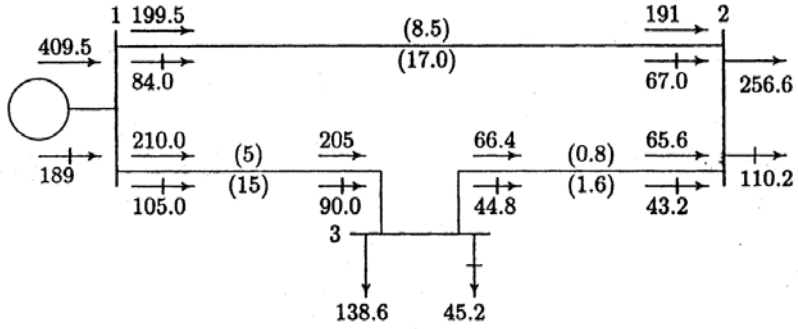
$$= 0.05 + j0.15 = 5.0\ MW + j15.0\ Mvar$$

$$\vec{S}_{L23} = \vec{S}_{23} + \vec{S}_{32} = -0.656 - j0.432 +$$

$$0.664 + j0.448$$

$$= 0.008 + j0.016 = 0.8\ MW + j1.6\ Mvar$$

- The power flow diagram is as shown, where the real power direction is indicated by ⟶ and the reactive power direction is indicated by ⟼.
- The values within parentheses are the real and reactive losses in the line.



Power flow diagram (powers in MW and Mvar)

- The total loss in the system is calculated by summing the loss over all the lines.

Modification of algorithm to include PV buses

- In the previous sections, we have assumed all buses are *PQ* buses, except the slack bus.
- In these buses, *P* and *Q* are specified and *V* and δ are to be determined.
- In PV buses, *P* and *V* as well as the limits on *Q* are specified and *Q* and δ are to be determined.
- Any bus where the voltage magnitude can be held constant at a specified value is called a *voltage controlled bus or generator bus* (there is a generator or compensator connected to the bus).
- The calculation strategy of load flow solution with PV buses is different for PQ buses.
- Let the bus be numbered as

$i = 1$ slack bus
$i = 2, 3, 4 \ldots, m$ PQ buses
$i = m+1, m+2, .., n$ PV buses

Modified computational procedure

- At the voltage controlled buses (PV buses), bus voltages are specified, i.e., $|\vec{V}_i| = |\vec{V}_i|_{spec}$; $Q_{i,min} < Q_i < Q_{i,max}$ or $Q_{Gi,min} < Q_{Gi} < Q_{Gi,max}$. Note $Q_{Gi} = Q_i + Q_{Di}$.
- The magnitude of the voltage has to be maintained at the specified value $|\vec{V}_i|_{spec}$.

- The voltage at the PV bus is controlled by controlling Q_i.
- Therefore, the values of Q_i and δ_i are to be updated in every iteration.
- For PV buses, only the angles δ_i^0 have to be assumed.
- Check type of buses.
  (a) If *i*th bus is a PQ bus, solve the voltage equation for bus *i* as usual.
  (b) If *i*th bus is PV bus, go to the next step.
  Set $|\vec{V}_i| = |\vec{V}_i|_{spec}$

Step 1

- Calculate the reactive power at the *i*th bus using the following equation:

$$Q_i = -Im[\vec{V}_i^* \sum_{j=1}^{n} \vec{Y}_{ij} \vec{V}_j] \quad i = m+1, \ldots, n$$

- The revised value of Q_i is obtained from above by substituting the most updated values of voltages on the right-hand side.
- For the (k+1)th iteration, we can write

$$Q_i^{k+1} = -Im[(\vec{V}_i^k)^* \sum_{j=1}^{i-1} \vec{Y}_{ij} \vec{V}_j^{k+1} + (\vec{V}_i^k)^* \sum_{j=i}^{n} \vec{Y}_{ij} \vec{V}_j^k]$$

for $i = m+1, \ldots, n$

- If the calculated reactive power is within limits, then this bus can be treated as PV bus and set $Q_i = Q_i^{k+1}$.

Step 2

- The revised value of δ_i is obtained immediately after step 1 using the following equation:

$$\delta_i^{k+1} = \angle \vec{V}_i^{k+1}$$

- Solve the voltage equation using (24) with

$$\vec{V}_i^{k+1} = \frac{1}{\vec{Y}_{ii}} \left[\frac{\vec{S}_i^*}{(\vec{V}_i^*)^k} - \sum_{j=1,}^{i-1} \vec{Y}_{ij}\, \vec{V}_j^{k+1} - \sum_{j=i+1}^{n} \vec{Y}_{ij}\, \vec{V}_j^{k} \right]$$

with $\vec{S}_i^* = P_i - jQ_i^{k+1}$

Violation of Q-limits

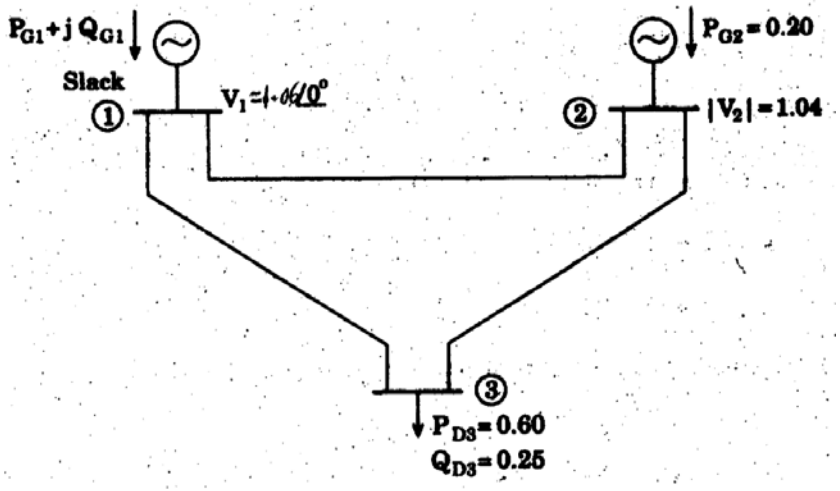
If the calculated reactive power violates the limits (the voltage cannot be maintained at the specified value due to lack of reactive power support), then this bus can be treated as PQ bus (bus specification is dropped) and set if

$$Q_i^{k+1} < Q_{i,min} \text{ , then } Q_i = Q_{i,min}$$
$$Q_i^{k+1} > Q_{i,max} \text{ , then } Q_i = Q_{i,max}$$

$\vec{V}_i^{k+1}$ is then calculated using these values, with a flat voltage start.

***Example*:** Determine the voltages of the various buses at the end of the first iteration. The voltage magnitude at bus 2 is to be maintained at 1.04 pu. The minimum and maximum reactive power limits of the generation at bus 2 are 0.0 and 0.3 pu, respectively, i.e., $0 < Q_{G2} < 0.3$. The bus admittance matrix is given by

$$\vec{Y}_{bus} = \begin{bmatrix} 6.67 - j19.89 & -1.67 + j5.0 & -5.0 + j15.0 \\ -1.67 + j5.0 & 4.17 - j12.4 & -2.5 + j7.5 \\ -5.0 + j15.0 & -2.5 + j7.5 & 7.5 - j22.39 \end{bmatrix}$$



The power injected into the network by each bus is

$$\vec{S}_1 = P_1 + jQ_1 \text{ , } \quad \vec{S}_2 = P_2 + jQ_2 = 0.2 + jQ_2$$

$$\vec{S}_3 = P_3 + jQ_3 = -0.6 - j0.25$$

- Assume $|\vec{V}_3^0| = 1$ (flat start) and $|\vec{V}_2^0| = |1.04 + j0|$, $\delta_2^0 = 0$, $|\vec{V}_2|_{spec} = 1.04$
- To find $\vec{V}_2^1$, first calculate Q_2^1 as follows:

$$Q_i^{k+1} = -Im[(\vec{V}_i^k)^* \sum_{j=1}^{i-1} \vec{Y}_{ij}\, \vec{V}_j^{k+1} + (\vec{V}_i^k)^* \sum_{j=i}^{n} \vec{Y}_{ij}\, \vec{V}_j^k]$$

For $i = 2$ and $n = 3$, we have

$$Q_2^{k+1} = -Im[(\vec{V}_2^k)^* \sum_{j=1}^{2-1} \vec{Y}_{2j}\, \vec{V}_j^{k+1} + (\vec{V}_2^k)^* \sum_{j=2}^{3} \vec{Y}_{2j}\, \vec{V}_j^k]$$

For the first iteration $k = 0$. Thus,

$$Q_2^1 = -Im[(\vec{V}_2^0)^* \vec{Y}_{21} \vec{V}_1^1 + (\vec{V}_2^0)^* \{\vec{Y}_{22} \vec{V}_2^0 + \vec{Y}_{23} \vec{V}_3^0\}]$$

$$= -Im[\,(\vec{V}_2^0)^* (\vec{Y}_{21} \vec{V}_1^1 + \vec{Y}_{22} \vec{V}_2^0 + \vec{Y}_{23} \vec{V}_3^0)]$$

$$= -Im[(1.04 - j0.0)\{(-1.67 + j5.0)(1.06 + j0.0) + (4.17 - j12.4)(1.04 + j0.0) + (-2.5 + j7.5)(1)\}]$$

$$= -\,I\,m\,[0.0695 - j0.09984] = 0.09984\ pu$$

- The value of Q_{G2}^1 $(Q_{G2}^1 = Q_2^1 + Q_{D2})$ with $Q_{D2} = 0$ is within the limits imposed by $Q_{G2,min} = 0.0 < Q_{G2} < Q_{G2,max} = 0.3$.
- Using this value of Q_2^1, we find $\delta_2^1 = \angle \vec{V}_2^1$.
- We use (26) with $\vec{S}_i^* = P_i - jQ_i^{k+1}$ to get

$$\vec{V}_2^1 = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^0} - \vec{Y}_{21}\vec{V}_1^1 - \vec{Y}_{23}\vec{V}_3^0\right] \text{ with } \vec{S}_2^* = P_2 - jQ_2^1$$

$$= \frac{1}{4.17 - j12.4}\left[\frac{0.2 - j0.09984}{1.04 - j0.0} - (-1.67 + j5.0)(1.06 + j0.0) - (-2.5 + j7.5)(1 + j0.0)\right]$$

$$= 1.043\angle 0.4985^o$$

This means $\delta_2^1 = 0.4985^o$ and $|\vec{V}_2^1| = |\vec{V}_2|_{spec} = 1.04$. Thus, $\vec{V}_2^1 = 1.04\angle 0.4985^o = 1.0399 + j0.009$.

To calculate $\vec{V}_3^1$ we use (26) with to get

$$\vec{V}_3^1 = \frac{1}{\vec{Y}_{33}}\left[\frac{P_3 - jQ_3}{(\vec{V}_3^*)^0} - \vec{Y}_{31}\vec{V}_1 - \vec{Y}_{32}\vec{V}_2^1\right]$$

$$= \frac{1}{7.5 - j22.39}\left[\frac{-0.6 + j0.25}{1 - j0.0} - (-5.0 + j15.0)(1.06 + j0.0) - (-2.5 + j7.5)(1.0399 + j0.009)\right]$$

$$= \frac{1}{7.5 - j22.39}(7.367 - j23.427)$$

$$= 1.0399 - j0.0192 = 1.04\angle -1.0577^o$$

-The bus voltages at the end of the first iteration are given by

$\vec{V}_1^1 = 1.06 + j0.0$, $\vec{V}_2^1 = 1.0399 + j0.009$,

$\vec{V}_3^1 = 1.0399 - j0.0192$

Example: Suppose in the previous example, the reactive power constraint on generator 2 is given as $0.2 < Q_{G2} < 0.5$. Find the values of $\vec{V}_2$ and $\vec{V}_3$ at the end of the first iteration.

- The value of Q_2^1 was earlier found to be 0.09984.
- $Q_{G2}^1 = Q_2^1 + Q_{D2} = 0.09984 + 0 = 0.09984.$
- This value violates the minimum reactive power generation of 0.2 and $\vec{V}_2$ is no longer fixed at 1.04 pu.
- Hence, the minimum reactive power generation $Q_{G2,min}$ is fixed at 0.2 pu and the PV bus is considered as a load bus (PQ bus).
- For bus 2, we have

$$P_2 = 0.2\ ,\ Q_2 = Q_{G2,min} - Q_{D2} = 0.2 - 0 = 0.2$$

- To start the iteration choose $\vec{V}_2^0 = \vec{V}_3^0 = 1\angle 0^o$.

$$\vec{V}_2^1 = \frac{1}{\vec{Y}_{22}}\left[\frac{\vec{S}_2^*}{(\vec{V}_2^*)^0} - \vec{Y}_{21}\vec{V}_1^1 - \vec{Y}_{23}\vec{V}_3^0\right], \quad \vec{S}_2 = P_2 + jQ_2$$

$$= \frac{1}{4.17-j12.4}\left[\frac{0.2-j0.2}{1} - (-1.67 + j5.0)(1.06 + j0.0) - (-2.5 + j7.5)(1 + j0.0)\right]$$

$$= \frac{1}{4.17-j12.4}(4.47 - j13.0) = 1.0508 + j0.00713$$

$$= 1.0508\angle 0.389^o$$

$$\vec{V}_3^1 = \frac{1}{\vec{Y}_{33}}\left[\frac{\vec{S}_3^*}{(\vec{V}_3^*)^0} - \vec{Y}_{31}\vec{V}_1 - \vec{Y}_{32}\vec{V}_2^1\right],$$

$$= \frac{1}{7.5-j22.39}\left[\frac{-0.6+j0.25}{1-j0.0} - (-5.0 + j15.0)(1.06 + j0.0) - (-2.5 + j7.5)(1.0508 + j0.00713)\right]$$

$$= \frac{1}{7.5-j22.39}(7.3804 - j23.513)$$

$$= 1.0435 - j0.0199 = 1.0437\angle -1.09^o$$

-These values are used to compute the new set of voltages in the next iteration.

Newton-Raphson method

- The Gauss-Seidel procedure converges very slowly toward a solution for systems with very large number of buses and in some cases even failing to converge.
- The Newton-Raphson method is normally used because of its fast-converging feature.

- To help us to understand the method, we will use a 2-bus system for illustration.
- Need to know the specified quantities $P_i, Q_i,\ i \neq 1$ so that the power flow equations $p_i, q_i,\ i \neq 1$ can be found in terms of the unknown, independent variables $\delta_i, V_i,\ i \neq 1$.
- For n = 2 and the slack bus voltage $\vec{V}_1 = 1\angle 0^o$, we need only write two equations for the real and reactive power at bus 2 as the number of unknowns = 2(n-1) – n_g = 2(2 - 1) – 0 = 2.
- Assume that P_2 and Q_2 are known.
- Then the two unknowns are δ_2 and V_2.

- Recall eqns. (18) and (19) as follows:

$$P_i = V_i \sum_{j=1}^{n} Y_{ij} V_j \cos(\delta_i - \delta_j - \gamma_{ij})$$

$$Q_i = V_i \sum_{j=1}^{n} Y_{ij} V_j \sin(\delta_i - \delta_j - \gamma_{ij})$$

- For n =2, we have

$$P_2 = V_2[Y_{21} V_1 \cos(\delta_2 - \delta_1 - \gamma_{21}) + Y_{22} V_2 \cos(\delta_2 - \delta_2 - \gamma_{22})] \quad \text{(R1)}$$

$$Q_2 = V_2[Y_{21} V_1 \sin(\delta_2 - \delta_1 - \gamma_{21}) + Y_{22} V_2 \sin(\delta_2 - \delta_2 - \gamma_{22})] \quad \text{(R2)}$$

- The equations (R1) and (R2) are nonlinear.
- Expanding the equations in Taylor series about the values of the unknown variables found at the *k*th iteration for the expressions for real power p and reactive power q at bus 2 with all higher terms neglected gives

$$P_2 = p_2(\delta_2^k, V_2^k) + \left[\frac{\partial p_2}{\partial \delta_2}\right]^k \Delta\delta_2^k + \left[\frac{\partial p_2}{\partial V_2}\right]^k \Delta V_2^k$$

$$Q_2 = q_2(\delta_2^k, V_2^k) + \left[\frac{\partial q_2}{\partial \delta_2}\right]^k \Delta\delta_2^k + \left[\frac{\partial q_2}{\partial V_2}\right]^k \Delta V_2^k \quad \text{(R3)}$$

- Rearranging terms and using matrix form yields

$$\begin{bmatrix}\Delta p_2 \\ \Delta q_2\end{bmatrix}^k = \begin{bmatrix}P_2 - p_2 \\ Q_2 - q_2\end{bmatrix}^k = \begin{bmatrix}\frac{\partial p_2}{\partial \delta_2} & \frac{\partial p_2}{\partial V_2} \\ \frac{\partial q_2}{\partial \delta_2} & \frac{\partial q_2}{\partial V_2}\end{bmatrix}^k \begin{bmatrix}\Delta\delta_2 \\ \Delta V_2\end{bmatrix}^k \quad \text{(R4)}$$

- Note that the lower case letters for p and q represent P and Q as given by (R1) and (R2).

- These lower-case letters tells us that these are expressions for which solutions are being sought but not yet attained.
- The upper-case letters indicate the specified values for *P* and *Q* at the bus.
- At the arrival of a solution for the voltage profile, the lower-case *p* and *q* take on the values of the upper-case *P* and *Q*.
- The subscript *k* on the matrices tells us that all quantities in the equation are evaluated at those values of the variable corresponding to the *k*th iteration.
- (R4) can be rewritten in a compact form:

$$\Delta\vec{\mathrm{u}}^{\mathrm{k}} = \mathbf{J}^{\mathrm{k}}\Delta\vec{\mathrm{x}}^{\mathrm{k}} \qquad \text{(R5)}$$

where

$$\Delta\vec{\mathrm{u}}^{\mathrm{k}} = \begin{bmatrix}\Delta p_2 \\ \Delta q_2\end{bmatrix}^{\mathrm{k}}, \quad \Delta\vec{\mathrm{x}}^{\mathrm{k}} = \begin{bmatrix}\Delta\delta_2 \\ \Delta \mathrm{V}_2\end{bmatrix}^{\mathrm{k}}, \; J^k = \begin{bmatrix}\frac{\partial p_2}{\partial \delta_2} & \frac{\partial p_2}{\partial V_2} \\ \frac{\partial q_2}{\partial \delta_2} & \frac{\partial q_2}{\partial V_2}\end{bmatrix}^{\mathrm{k}}$$

- (R5) is called the *mismatch equation* as Δp_2 and Δq_2 is the real power mismatch and reactive power mismatch, respectively.
- J is called the *Jacobian* matrix.
-Now from (R5), $\Delta\vec{\mathrm{x}}^{\mathrm{k}}$ is given by

$$\Delta\vec{\mathrm{x}}^{\mathrm{k}} = (\mathbf{J}^{-1})^{\mathrm{k}}\Delta\vec{\mathrm{u}}^{\mathrm{k}} \qquad \text{(R6)}$$

- The vector $\vec{\mathrm{x}}$ is called the state vector of the electric power system.
- The updated values of the system state vector is

$$\vec{\mathrm{x}}^{\mathrm{k+1}} = \vec{\mathrm{x}}^{\mathrm{k}} + \Delta\vec{\mathrm{x}}^{\mathrm{k}} \qquad \text{(R7)}$$

where

$$\delta_{\mathrm{i}}^{\mathrm{k}} = \delta_{\mathrm{i}}^{\mathrm{k-1}} + \Delta\delta_{\mathrm{i}}^{\mathrm{k-1}} \qquad \text{(R8)}$$

$$\mathrm{V}_{\mathrm{i}}^{\mathrm{k}} = \mathrm{V}_{\mathrm{i}}^{\mathrm{k-1}} + \Delta\mathrm{V}_{\mathrm{i}}^{\mathrm{k-1}} \qquad \text{(R9)}$$

-The iteration process stops when Δp_2^k and Δq_2^k are within tolerance, i.e., $|\Delta p_2^k| \leq \epsilon, \; |\Delta q_2^k| \leq \epsilon$.

- By applying (18) and (19) at bus 2 and using $\vec{V}_1 = V_1\angle\delta = 1\angle 0^o$ gives

$$p_2 = Y_{21}V_2\cos(\delta_2 - \gamma_{21}) + Y_{22}V_2^2\cos\gamma_{22} \qquad \text{(R10)}$$

$$q_2 = Y_{21}V_2\sin(\delta_2 - \gamma_{21}) - Y_{22}V_2^2\sin\gamma_{22} \qquad \text{(R11)}$$

- Now, we use the lower-case letter for *p* and *q* when preparing to perform an iteration procedure in search of the quantities on the right-hand side that make p_2 and q_2 equal to the *specified values*, respectively.

- (R4) makes clear the need to evaluate the partial derivatives of each power flow equation wrt each of the unknown variables in the system.
- Now the partial differentiation of p_2 and q_2 in (R10) and (R11) yields the following:

$$\frac{\partial p_2}{\partial \delta_2} = -Y_{21}V_2 \sin(\delta_2 - \gamma_{21}) \qquad \text{(R12)}$$

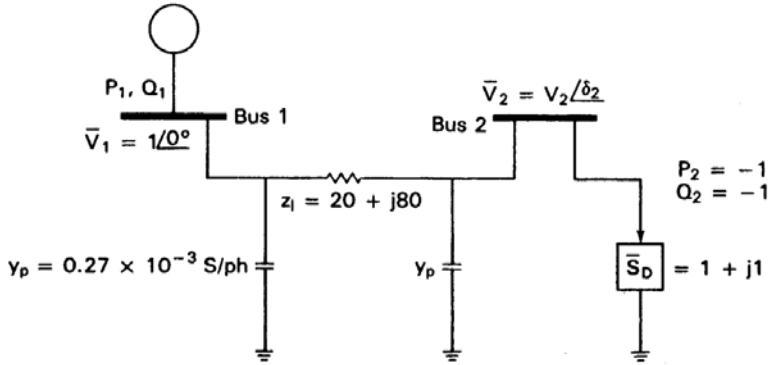
$$\frac{\partial p_2}{\partial V_2} = Y_{21} \cos(\delta_2 - \gamma_{21}) + 2Y_{22}V_2 \cos\gamma_{22} \qquad \text{(R13)}$$

$$\frac{\partial q_2}{\partial \delta_2} = Y_{21}V_2 \cos(\delta_2 - \gamma_{21}) \qquad \text{(R14)}$$

$$\frac{\partial q_2}{\partial V_2} = Y_{21} \sin(\delta_2 - \gamma_{21}) - 2Y_{22}V_2 \sin\gamma_{22} \qquad \text{(R15)}$$

- With the elements of $\vec{Y}_{bus}$ known, the solution procedure can then be carried out.

Example: A two-bus system is selected to illustrate the solution procedure to find the phasor voltage at bus 2, i.e., $\vec{V}_2$. The specific two-bus system to be analyzed here for power flow along the transmission line is as shown. Note that P_2 and Q_2 are specified.



- The bus admittance matrix for the two-bus transmission network is

$$\vec{Y}_{bus} = \begin{bmatrix} 6.272\angle -75.683^o & 6.4105\angle 104^o \\ 6.4105\angle 104^o & 6.272\angle -75.683^o \end{bmatrix}$$

- Thus, from (R1) and (R2),

$$p_2 = Y_{21}V_2 \cos(\delta_2 - \gamma_{21}) + Y_{22}V_2^2 \cos\gamma_{22}$$

$$= 6.4105V_2 \cos(\delta_2 - 104^o) +$$

$$6.272V_2^2 \cos(-75.683^o)$$

$$= 6.4105V_2 \cos(\delta_2 - 104^o) + 1.551V_2^2 \qquad \text{(R16)}$$

$$q_2 = Y_{21}V_2\sin(\delta_2 - \gamma_{21}) - Y_{22}V_2^2\sin\gamma_{22}$$
$$= 6.4105V_2\sin(\delta_2 - 104^o) -$$
$$6.272V_2^2\sin(-75.683^o)$$
$$= 6.4105V_2\sin(\delta_2 - 104^o) + 6.0772V_2^2 \quad \text{(R17)}$$

-Similarly, for the Jacobian,

$$\frac{\partial p_2}{\partial \delta_2} = -6.4105V_2\sin(\delta_2 - 104^o) \quad \text{(R18)}$$

$$\frac{\partial p_2}{\partial V_2} = 6.4105\cos(\delta_2 - 104^o) + 3.102V_2 \quad \text{(R19)}$$

$$\frac{\partial q_2}{\partial \delta_2} = 6.4105V_2\cos(\delta_2 - 104^o) \quad \text{(R20)}$$

$$\frac{\partial q_2}{\partial V_2} = 6.4105\sin(\delta_2 - 104^o) + 12.1544V_2 \quad \text{(R21)}$$

Solution procedure

-The purpose is to apply the NR method in finding a solution for V_2 and δ_2 so that these values will satisfy (R16) and (R17) for the P_2 and Q_2 *values specified*.

- The process is begun by assuming a *flat profile*, i.e., set all magnitudes of the unknown bus voltages equal to unity and all the phase angles equal to 0°.

- With $\vec{V}_2 = 1\angle 0^o$, we have from (R16), (R17) and (R18) – (R21)

$$p_2^0 = 6.4105V_2^0\cos(\delta_2^0 - 104^o) + 1.551(V_2^0)^2$$
$$= 6.4105(1)\cos(0^o - 104^o) + 1.551(1)^2$$
$$= 0.0002$$

$$q_2^0 = 6.4105V_2^0\sin(\delta_2^0 - 104^o) + 6.0772(V_2^0)^2$$
$$= 6.4105(1)\sin 0^o - 104^o) + 6.0772(1)^2$$
$$= -0.1429$$

$$\Delta\vec{u}^0 = \begin{bmatrix}\Delta p_2^0 \\ \Delta q_2^0\end{bmatrix} = \begin{bmatrix}P_2 - p_2^0 \\ Q_2 - q_2^0\end{bmatrix} = \begin{bmatrix}-1 - 0.00016 \\ -1 + 0.1429\end{bmatrix}$$
$$= \begin{bmatrix}-1.0002 \\ -0.8571\end{bmatrix}$$

-Find elements of the Jacobian

$$\left[\frac{\partial p_2}{\partial \delta_2}\right]^0 = -6.4105V_2^0\sin(\delta_2^0 - 104^o)$$
$$= -6.4105(1)\sin(0^o - 104^o) = 6.221$$

$$\left[\frac{\partial p_2}{\partial V_2}\right]^0 = 6.4105\cos(\delta_2^0 - 104^o) + 3.102V_2^0$$
$$= 6.4105\cos(0^o - 104^o) + 3.102(1)$$
$$= 1.5512$$

$$\left[\frac{\partial q_2}{\partial \delta_2}\right]^0 = 6.4105V_2^0\cos(\delta_2^0 - 104^o)$$
$$= 6.4105(1)\cos(0^o - 104^o) = -1.5508$$

$$\left[\frac{\partial q_2}{\partial V_2}\right]^0 = 6.4105\sin(\delta_2^0 - 104^o) + 12.1544V_2^0$$
$$= 6.4105\sin(0^o - 104^o) + 12.1544(1)$$
$$= 5.9343$$

$$\Rightarrow \mathbf{J}^0 = \begin{bmatrix} \frac{\partial p_2}{\partial \delta_2} & \frac{\partial p_2}{\partial V_2} \\ \frac{\partial q_2}{\partial \delta_2} & \frac{\partial q_2}{\partial V_2} \end{bmatrix}^0 = \begin{bmatrix} 6.221 & 1.5512 \\ -1.5508 & 5.9343 \end{bmatrix}^0$$

-From (R6),

$$\Delta\vec{x}^0 = \begin{bmatrix} \Delta\delta_2^0 \\ \Delta V_2^0 \end{bmatrix} = (\mathbf{J}^{-1})^0\Delta\vec{u}^0$$
$$= \begin{bmatrix} 0.1509 & -0.0395 \\ 0.0394 & 0.1582 \end{bmatrix}\begin{bmatrix} -1.0002 \\ -0.8571 \end{bmatrix}$$
$$= \begin{bmatrix} -0.117 \text{ rad} \\ -0.1751 \end{bmatrix} = \begin{bmatrix} -6.712^o \\ -0.1751 \end{bmatrix}$$

Note: The angle is in radians (convert to degrees).

1st iteration

-The first corrected values of the unknown variables are (see (R8) and (R9))

$$\delta_2^1 = \delta_2^0 + \Delta\delta_2^0 = 0^o - 6.712^o = -6.712^o$$
$$V_2^1 = V_2^0 + \Delta V_2^0 = 1 - 0.175 = 0.8249 \text{ pu}$$

-Using these values in (R16) and (R17) gives

$$p_2^1 = 6.4105\,V_2^1\cos(\delta_2^1 - 104^o) + 1.551(V_2^1)^2$$
$$= (6.4105)(0.8249)\cos(-6.712^o - 104^o) +$$
$$1.551(0.8249)^2 = -0.8148$$
$$q_2^1 = 6.4105V_2^1\sin(\delta_2^1 - 104^o) + 6.0772(V_2^1)^2$$
$$= (6.4105)(0.8249)\sin(-6.712^o - 104^o) +$$
$$6.0772(0.8429)^2 = -0.8109$$

2nd iteration

$$\Delta\vec{u}^1 = \begin{bmatrix} \Delta P_2^1 \\ \Delta Q_2^1 \end{bmatrix} = \begin{bmatrix} P_2 - p_2^1 \\ Q_2 - q_2^1 \end{bmatrix} = \begin{bmatrix} -1 + 0.8148 \\ -1 + 0.8109 \end{bmatrix}$$
$$= \begin{bmatrix} -0.1852 \\ -0.1891 \end{bmatrix}$$

-Find elements of the Jacobian

$$\left[\frac{\partial p_2}{\partial \delta_2}\right]^1 = -6.4105 V_2^1 \sin(\delta_2^1 - 104^o)$$
$$= -6.4105(0.8249)\sin(-6.712^o - 104^o)$$
$$= 4.9467$$

$$\left[\frac{\partial p_2}{\partial V_2}\right]^1 = 6.4105\cos(\delta_2^1 - 104^o) + 3.102\, V_2^1$$
$$= 6.4105\cos(-6.712^o - 104^o)$$
$$+ 3.102(0.8249) = 0.2917$$

$$\left[\frac{\partial q_2}{\partial \delta_2}\right]^1 = 6.4105 V_2^1 \cos(\delta_2^1 - 104^o)$$
$$= 6.4105(0.8249)\cos(-6.712^o - 104^o)$$
$$= -1.8702$$

$$\left[\frac{\partial q_2}{\partial V_2}\right]^1 = 6.4105\sin(\delta_2^1 - 104^o) + 12.1544 V_2^1$$
$$= 6.4105\sin(-6.712^o - 104^o) +$$
$$12.1544(0.8249) = 4.0301$$

$$\Rightarrow \mathbf{J}^1 = \begin{bmatrix} \frac{\partial p_2}{\partial \delta_2} & \frac{\partial p_2}{\partial V_2} \\ \frac{\partial q_2}{\partial \delta_2} & \frac{\partial q_2}{\partial V_2} \end{bmatrix}^1 = \begin{bmatrix} 4.9463 & 0.2917 \\ -1.8702 & 4.0301 \end{bmatrix}^1$$

$$\Delta\vec{x}^1 = \begin{bmatrix} \Delta\delta_2^1 \\ \Delta V_2^1 \end{bmatrix} = (\mathbf{J}^{-1})^1 \Delta\vec{u}^1$$

$$= \begin{bmatrix} 0.1968 & -0.0142 \\ 0.0913 & 0.2415 \end{bmatrix}\begin{bmatrix} -0.1852 \\ -0.1891 \end{bmatrix}$$

$$= \begin{bmatrix} -0.0338 \text{ rad} \\ -0.0626 \end{bmatrix} = \begin{bmatrix} -1.934^o \\ -0.0626 \end{bmatrix}$$

and

$$\delta_2^2 = \delta_2^1 + \Delta\delta_2^1 = -6.712^o - 1.923 = -8.646^o$$

$$V_2^2 = V_2^1 + \Delta V_2^1 = 0.8249 - 0.0626 = 0.7623 \text{ pu}$$

- The process is continued and a solution is converged in five iterations as given in the table below.

Iteration	δ_2(in deg)	V_2
0	0	1
1	-6.712°	0.8249
2	-8.646 °	0.7623
3	-8.9089 °	0.7533
4	-8.9143 °	0.7531
5	-8.9143 °	0.7531

NR for n-bus system

- For an n-bus system, the iteration formula is given by

$$\begin{bmatrix}\Delta P\\ \Delta Q\end{bmatrix}^k = \begin{bmatrix}J_1 & J_2\\ J_3 & J_4\end{bmatrix}^k \begin{bmatrix}\Delta\delta\\ \Delta V\end{bmatrix}^k$$

where

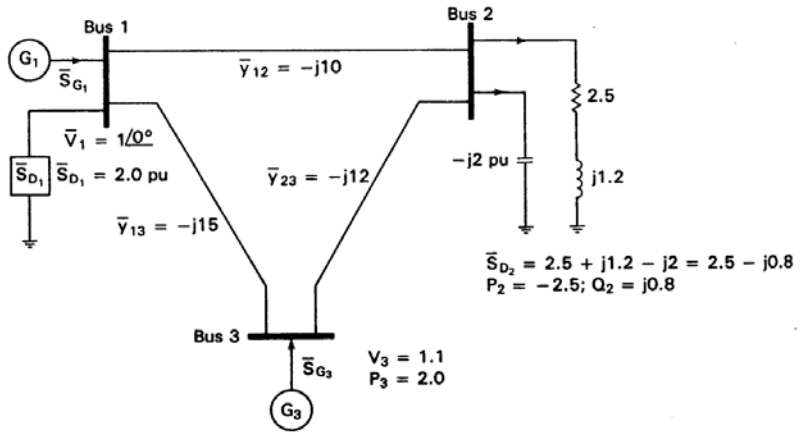
$$\Delta P = \begin{bmatrix}\Delta p_2^k\\ \vdots\\ \Delta p_n^k\end{bmatrix}, \Delta Q = \begin{bmatrix}\Delta q_2^k\\ \vdots\\ \Delta q_n^k\end{bmatrix}, \Delta\delta = \begin{bmatrix}\Delta\delta_2^k\\ \vdots\\ \Delta\delta_n^k\end{bmatrix}, \Delta V = \begin{bmatrix}\Delta V_2^k\\ \vdots\\ \Delta V_n^k\end{bmatrix}$$

$$J_1 = \begin{bmatrix}\left[\frac{\partial p_2}{\partial\delta_2}\right]^k & \cdots & \left[\frac{\partial p_2}{\partial\delta_n}\right]^k\\ \vdots & \ddots & \vdots\\ \left[\frac{\partial p_n}{\partial\delta_2}\right]^k & \cdots & \left[\frac{\partial p_n}{\partial\delta_n}\right]^1\end{bmatrix}, J_2 = \begin{bmatrix}\left[\frac{\partial p_2}{\partial V_2}\right]^k & \cdots & \left[\frac{\partial p_2}{\partial V_n}\right]^k\\ \vdots & \ddots & \vdots\\ \left[\frac{\partial p_n}{\partial V_2}\right]^k & \cdots & \left[\frac{\partial p_n}{\partial V_n}\right]^1\end{bmatrix}$$

$$J_3 = \begin{bmatrix}\left[\frac{\partial q_2}{\partial\delta_2}\right]^k & \cdots & \left[\frac{\partial q_2}{\partial\delta_n}\right]^k\\ \vdots & \ddots & \vdots\\ \left[\frac{\partial q_n}{\partial\delta_2}\right]^k & \cdots & \left[\frac{\partial q_n}{\partial\delta_n}\right]^1\end{bmatrix}, J_4 = \begin{bmatrix}\left[\frac{\partial q_2}{\partial V_2}\right]^k & \cdots & \left[\frac{\partial q_2}{\partial V_n}\right]^k\\ \vdots & \ddots & \vdots\\ \left[\frac{\partial q_n}{\partial V_2}\right]^k & \cdots & \left[\frac{\partial q_n}{\partial V_n}\right]^1\end{bmatrix}$$

- Need to know the specified quantities $P_i, Q_i,\ i \neq 1$ so that the power flow equations $p_i, q_i,\ i \neq 1$ can be found in terms of the unknown, independent variables $\delta_i, V_i,\ i \neq 1$.

Example: Consider the following three-bus system as shown.



- Bus1 is a slack bus, bus 2 is a load bus and bus 3 is a generator bus.
- P_2, Q_2 and P_3 (also V_3) are specified (known).
- The number of unknowns is 2(n-1) – n_g = 2(3-1) – 1 = 3.
- The three unknowns are δ_2, δ_3 and V_2.
- The power flow equations are given by

$$p_2 = V_2 \sum_{j=1}^{3} Y_{2j} V_j \cos(\delta_2 - \delta_j - \gamma_{2j})$$

$$q_2 = V_2 \sum_{j=1}^{3} Y_{2j} V_j \sin(\delta_2 - \delta_j - \gamma_{2j})$$

$$p_3 = V_3 \sum_{j=1}^{3} Y_{3j} V_j \cos(\delta_3 - \delta_j - \gamma_{3j})$$

- With these, one can find the Jacobian matrix.

Summary

The iterative methods mainly used to solve power flow problems are the Gauss-Seidel method, the Newton-Raphson method and the fast decoupled power flow method (not covered).

NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING

EE4530
Power System Analysis and Control
AY 2025/2026

Dr Foo Yi Shyh Eddy
Office: S1-B1c-89
Tel: 6790-4519

E-mail: eddyfoo@ntu.edu.sg

Acknowledgement: Part of this lecture series are based on materials obtained from Dr Wang Youyi, whose help is hereby gratefully acknowledged. Revised July 2025.

EE4530 - Power System Analysis and Control

Teaching Schedule

- Active Power, Frequency Control and Economic Dispatch (8 hours, Week 3 - Week 6)

Textbook

[1] Grainger, John J and Stevenson, William D, Jr, Power System Analysis, McGraw-Hill, 1994.

References

[1] Bergen A R and Vittal V, Power System Analysis, 2nd Edition, Prentice Hall, 2000.
[2] Weedy B M and Cory B J, Electric Power Systems, 4th Edition, John Wiley & Sons, 1998.
[3] Hingorani N G and Gyugyi L, Understanding FACTS: Concepts and Technology of Flexible AC Transmission Systems, IEEE Press, 2000.

Active Power and Frequency Control (8 hours)

Governor control systems.
Area control errors.
Load frequency control.
Transmission losses, penalty factors and loss coefficients.
Economic dispatch.
Automatic generation control.
Active power control devices.

Chapter 2
Active Power and Frequency Control

2.1 Introduction

The main objective of power system operation and control is to maintain continuous supply of power with an acceptable quality, to all the consumers in the system. The system will be in equilibrium, when there is a balance between the power demand and the power generated. As the power in AC form has real and reactive components: the real power balance; as well as the reactive power balance is to be achieved.

The function of an electric power system is to convert energy from one of the naturally available forms to the electrical form and to transfer it to the points of consumption. A properly designed and operated power system should meet the following fundamental requirements:

1. The system must be able to meet the continually changing load demand for both active and reactive power.

2. The system should support energy at the minimum cost and with the minimum ecological impact.

3. The "quality" of power supply must meet certain minimum standards with regard to the following factors:
   (a) Constancy of frequency;
   (b) Constancy of voltage; and
   (c) Level of reliability.

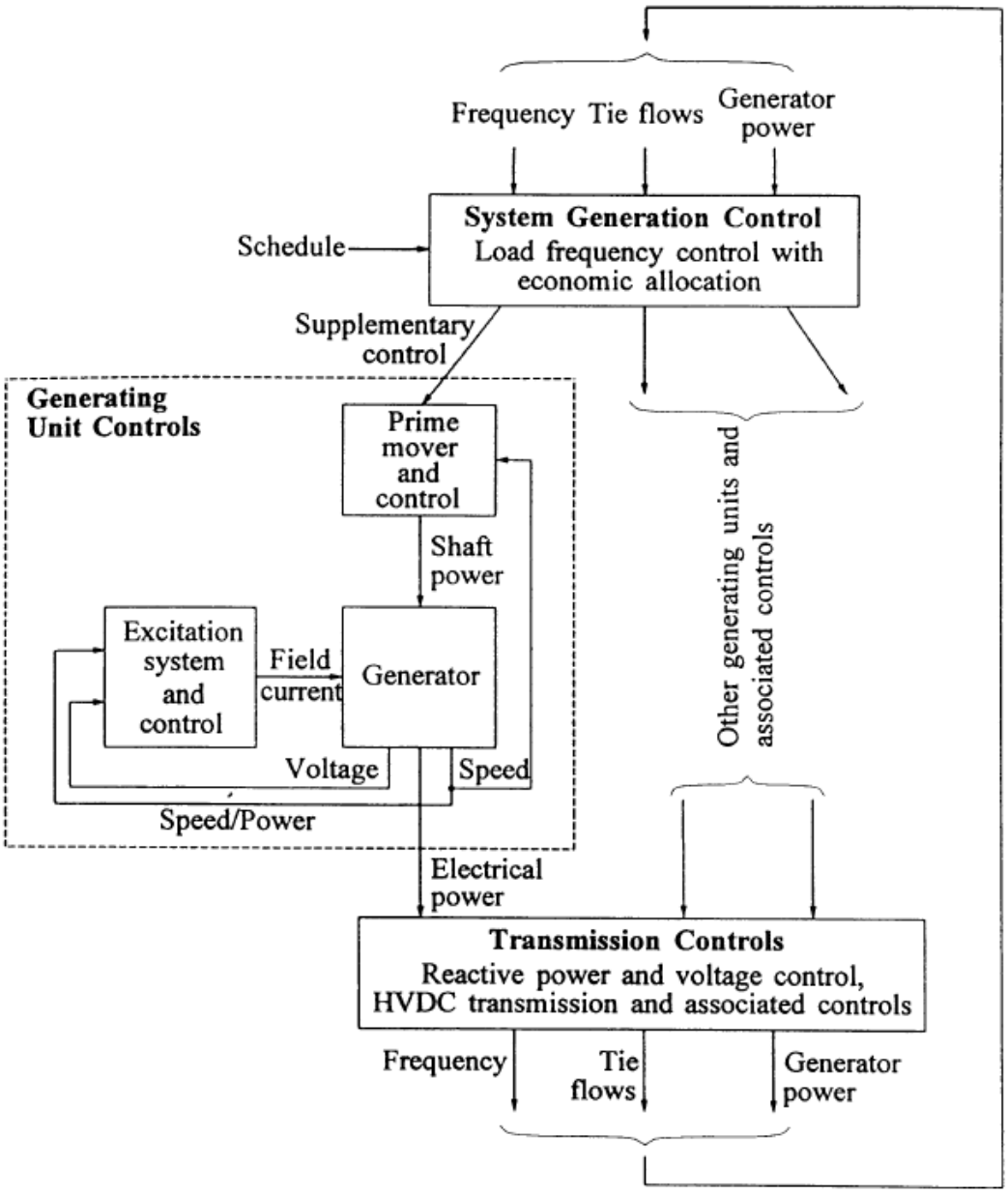


Figure 2.1 Substation of a power system and associated controls

Figure 2.1 shows the various substations of a power system and the associated controls. In this overall structure, there are controllers operating directly on individual system elements. In a generating unit these consist of prime mover controls and generator excitation controls. The prime mover controls are concerns with speed regulation and control of energy supply system variables such as boiler pressures, temperatures, and flows. The function of the generator excitation controls is to regulate generator voltage and reactive power output. The desired MW outputs of the individual generating unit are determined by the system generation control.

The primary purpose of the system generation controls is to balance the total system generation against loads and losses so that the desired frequency and power interchange with the neighbouring systems (tie flows) are maintained.

Figure 2.2 show an interconnected power network. Area-i is interconnected with its neighbouring systems Area-j and Area-k by the tie lines.

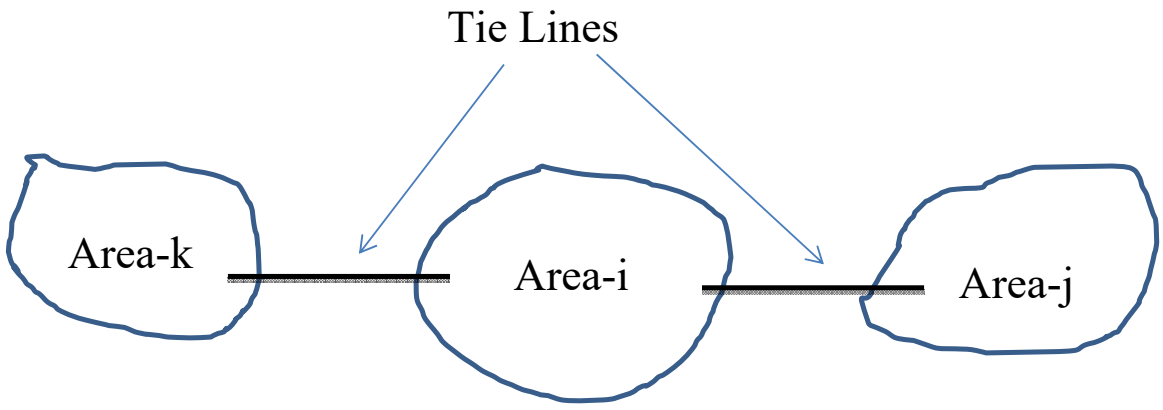


Figure 2.2 An interconnected power network.

Figure 2.3 shows the structure of a power network control.

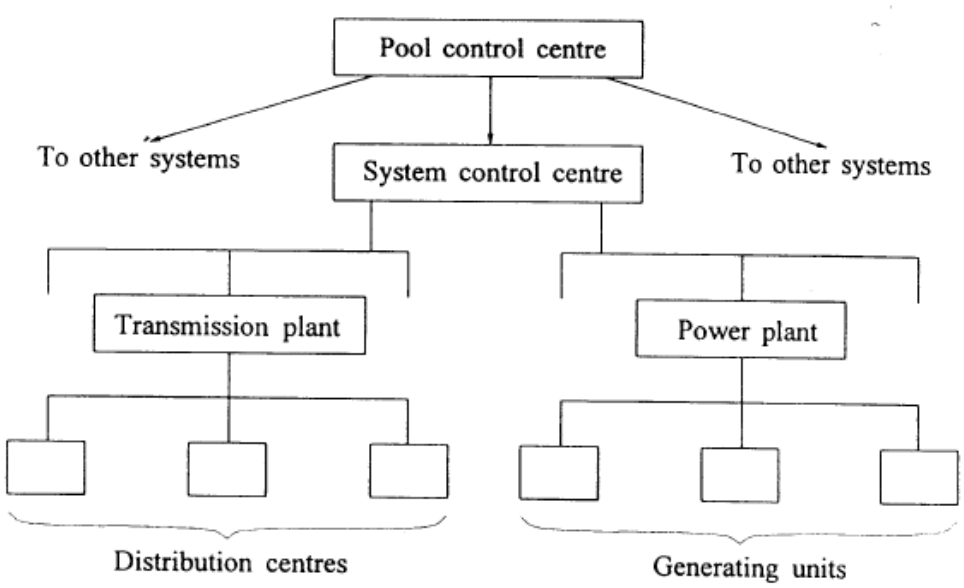


Figure 2.3 Structure of a power network control

In this structure, there are controllers operating directly on individual system elements such as excitation systems, prime movers, boilers, etc. There is usually some form of overall plant controller that coordinates the controls of closely linked elements. The plant controllers are in turn supervised by system controllers at the operating centers. The system controller actions are coordinated by pool-level master controllers. The overall control system is thus highly distributed, and relies on many different types of telemetering and control signals. Supervisory Control and Data Acquisition (SCADA) systems provide information to indicate the system working status.

Figure 2.4 shows the typical function performed at an energy control center and the diagram showing the data collection from a power system is shown in Figure 2.5.

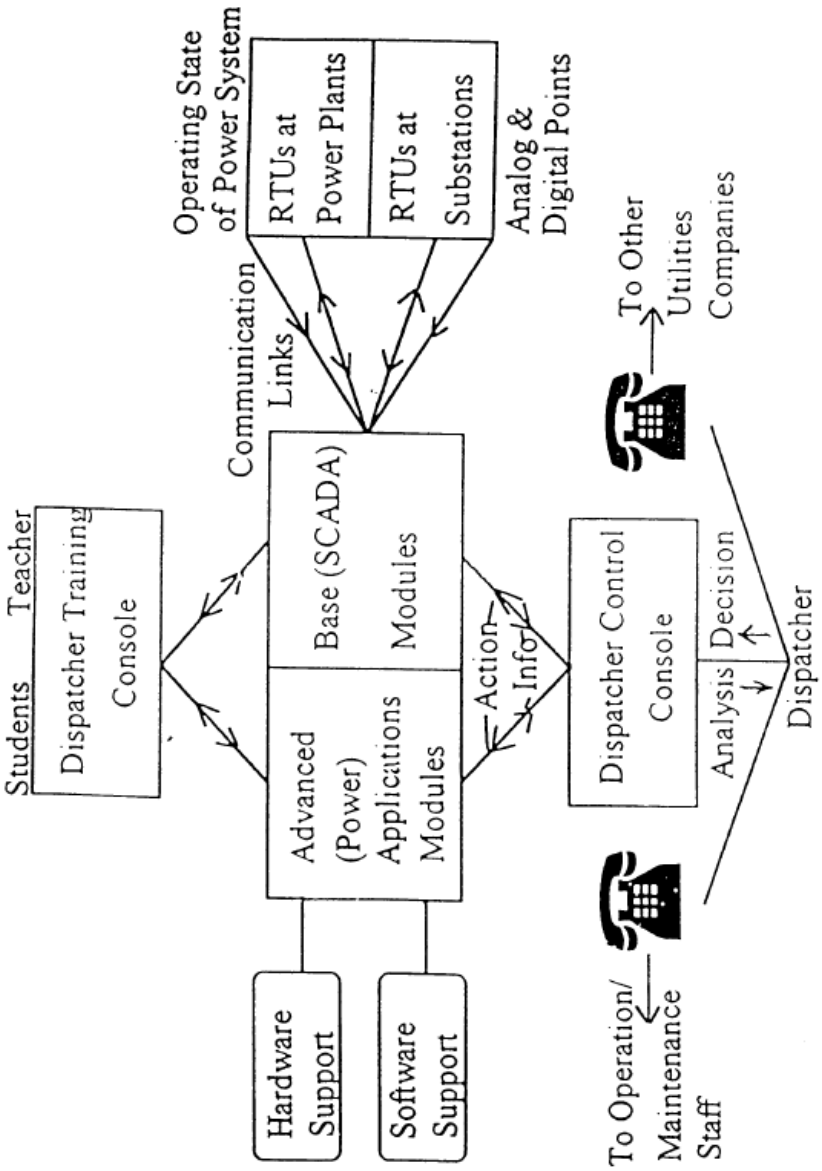


Figure 2.4 Typical function performed at an energy control center

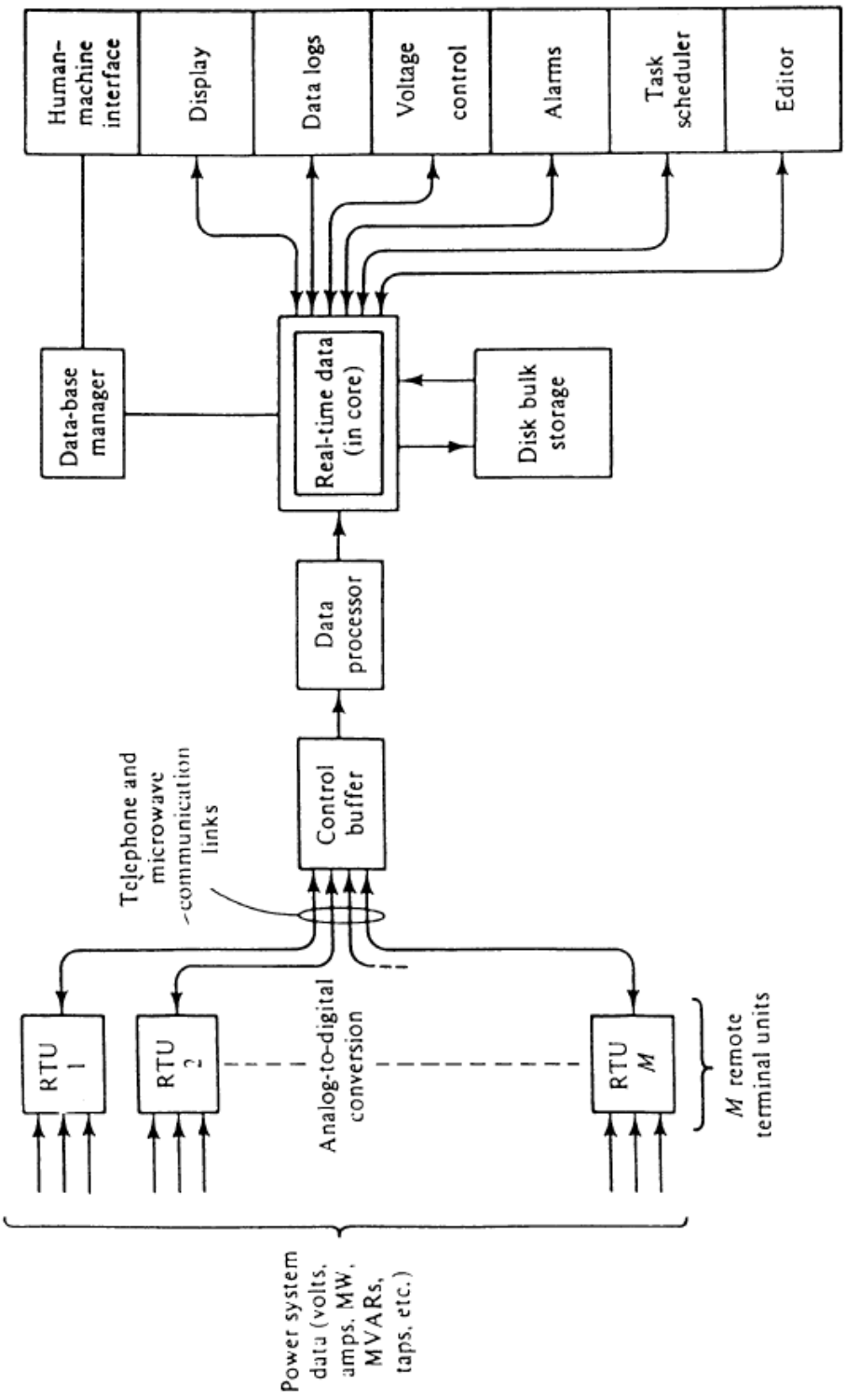


Figure 2.5 Data collection from a power system

The overview diagram of power application functions is given in Figure 2.6.

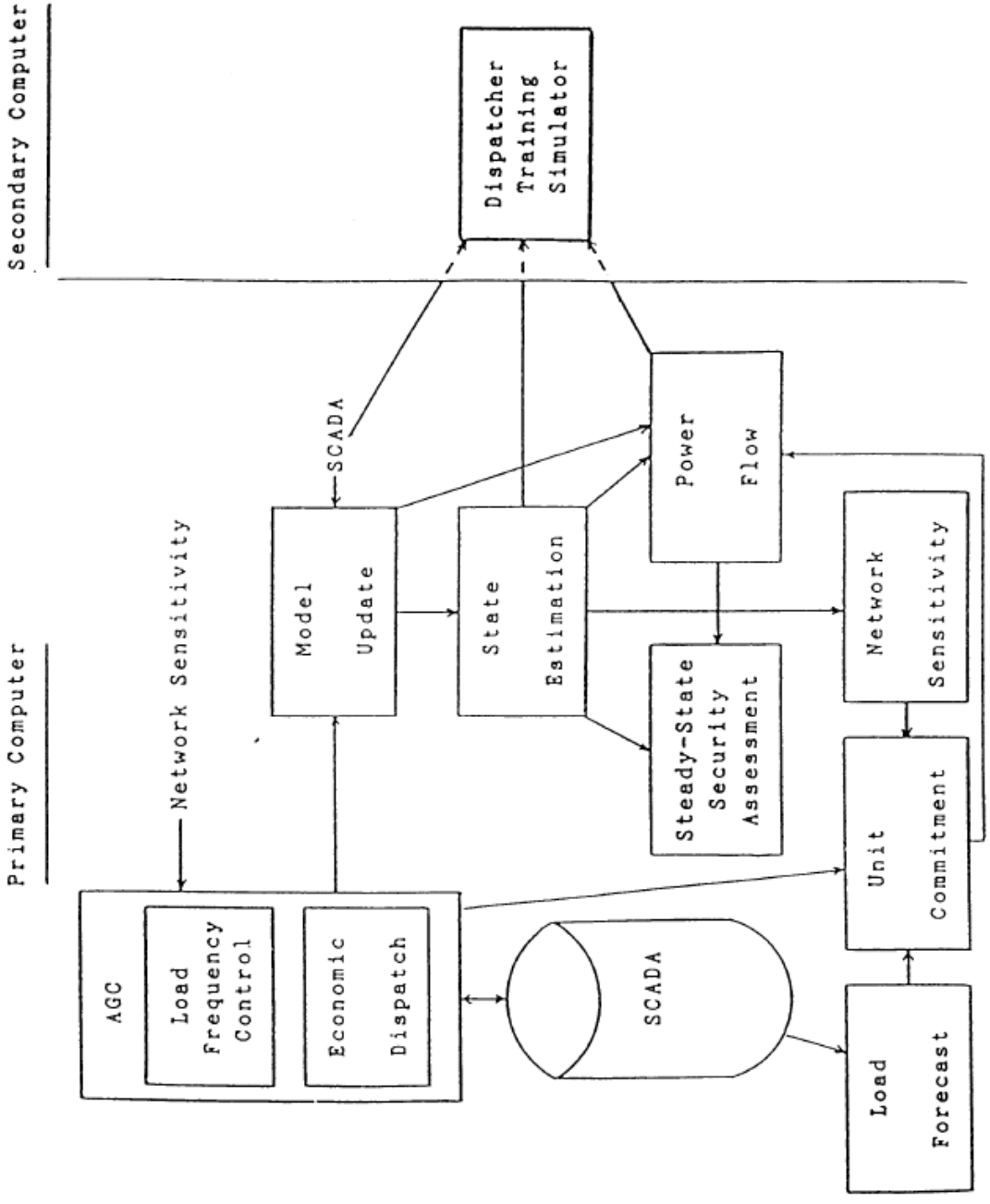


Figure 2.6 Overview diagram of power application functions

2.2 Automatic Generation Control

The load in a power system always changes. Changes in real (active) power affect mainly the system frequency, while reactive power is less sensitive to changes in frequency and is mainly dependent on changes in voltage magnitudes. Thus, real and reactive powers are controlled separately. The ***load frequency control*** (LFC) loop controls the real power and frequency and the ***automatic voltage regulator*** (AVR) loop regulates the reactive power and voltage magnitude. Load frequency control has gained in importance with the growth of interconnected systems and has made the operation of interconnected system possible.

The role of ***automatic generation control*** (AGC) is to control the tie line power under normal operating conditions of a power system.

2.2.1 Basic Generator Control Loops

In an interconnected power system, load frequency control and automatic voltage regulator equipment are installed for each generator. Figure 2.7 represents the schematic diagram of the load frequency control loop and automatic voltage regulator loop.

The controllers are set for a particular operating condition and take care of small changes in load demand to maintain the frequency and voltage magnitude within the specified limits. Small changes in real power are mainly dependent on changes in rotor angle δ and, thus, the frequency, f. The reactive power is mainly dependent on voltage magnitude (i.e. on the

generator excitation). The excitation system time constant is much smaller than the prime mover time constant and its transient decay much faster and does not affect the LFC dynamic.

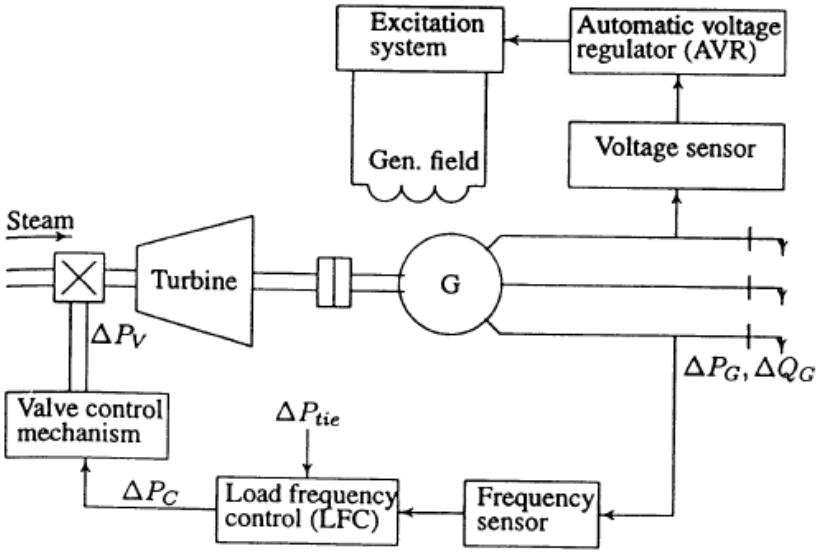


Figure 2.7 Schematic diagram of LFC and AVR of a generator

2.2.2 Load Frequency Control

A two-area interconnected power network is shown in Figure 2.8.

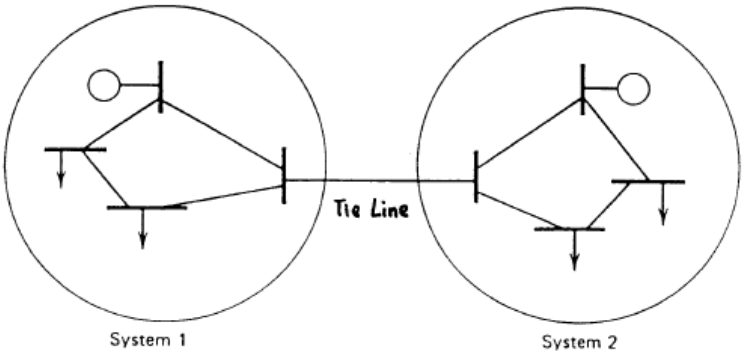


Figure 2.8 A two-area interconnected power network

The operation objectives of the LFC are to maintain reasonably uniform frequency, to divide the load between generators, and to control the tie-line interchange schedules. The change in frequency and tie-line real power are sensed, which is a measure of the change in rotor angle, δ, i.e., the error $\Delta\delta$ to be corrected. The error signals, i.e., Δf and ΔP_{tie}, are used to control the prime mover. The single area system diagram for LFC is given in Figure 2.9.

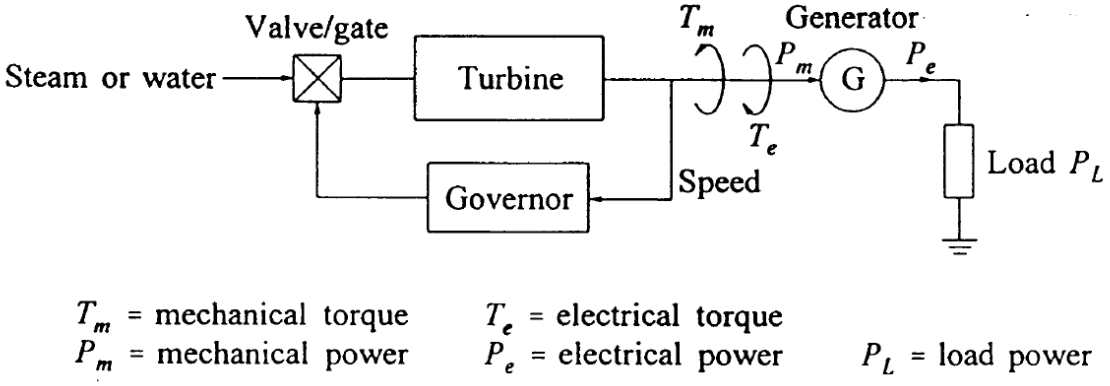


Figure 2.9 Single area system diagram for LFC

GENERATOR MODEL:

Consider the small deviation in the speed of a generator, we have

$$\frac{d\Delta\frac{\omega}{\omega_s}}{dt} = \frac{1}{2H}(\Delta P_m - \Delta P_e) \tag{2.1}$$

where $\omega = 2\pi f$, f denotes the frequency, in radian. $\omega_s = 2\pi f_s$, f_s = 50 Hz; H is the per unit inertia constant, in seconds, P_m

and P_e are the per unit mechanical and electrical power, respectively.

With speed expressed in per unit, we have

$$\frac{d\Delta\omega}{dt} = \frac{1}{2H}(\Delta P_m - \Delta P_e) \tag{2.2}$$

Taking Laplace transform gives that

$$\Delta\Omega(s) = \frac{1}{2Hs}(\Delta P_m(s) - \Delta P_e(s)) \tag{2.3}$$

Figure 2.10 shows the block diagram of the generator

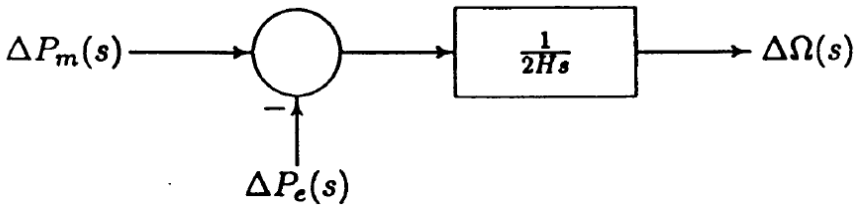


Figure 2.10 Generator block diagram

LOAD MODEL:

The load on a power system consists of a variety of electrical devices. For resistive loads, such as lighting, and heating loads, the electrical power is independent of frequency. However, a lot of loads, such as motors, are sensitive to changes in frequency. The speed-load characteristic of a composite load is approximated by

$$\Delta P_m = \Delta P_e = \Delta P_L + D\Delta\omega \tag{2.4}$$

NO losses in generator

where ΔP_L is the nonfrequency-sensitive load change, and $D\Delta\omega$ is the frequency-sensitive load change. D expressed as percent change in load divided by percent change in frequency. For example, if a load is changed by 1.6% for a 1% change in frequency, then D = 1.6. The generator block diagram with the load model results in the block diagram of Figure 2.11.

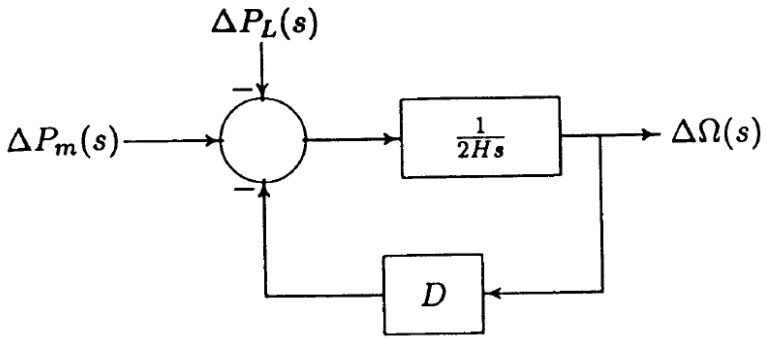


Figure 2.11 Generator and load block diagram

Figure 2.12 shows that the generator and load block diagram eliminating the simple feedback loop.

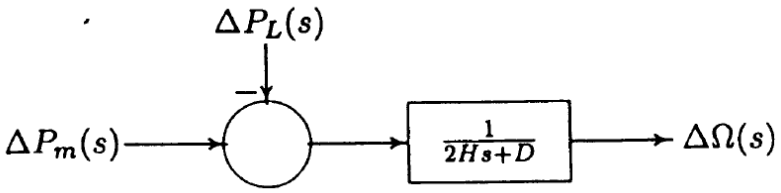


Figure 2.12 Generator and load block diagram

PRIME MOVER MODEL:

The source of mechanical power, commonly known as the prime mover, may be hydraulic turbines, steam turbines and gas turbines. The model of the turbine relates changes in mechanical power output ΔP_m to changes in valve position ΔP_v. Different types of turbines vary widely in characteristics. The simplest prime mover model for the nonreheat steam turbine is

$$G_T(s) = \frac{\Delta P_m(s)}{\Delta P_v(s)} = \frac{1}{1+\tau_T s} \tag{2.5}$$

where τ_T is the time constant, which is the range of 0.2 to 2.0 seconds.

Figure 2.13 show the block diagram.

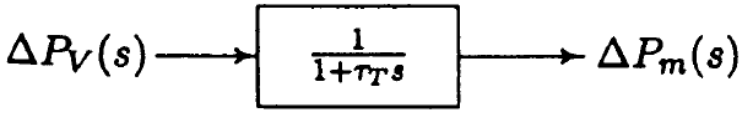


Figure 2.13 Nonreheat steam turbine block diagram

GOVERNOR MODEL:

Figure 2.14 shows a speed governing system

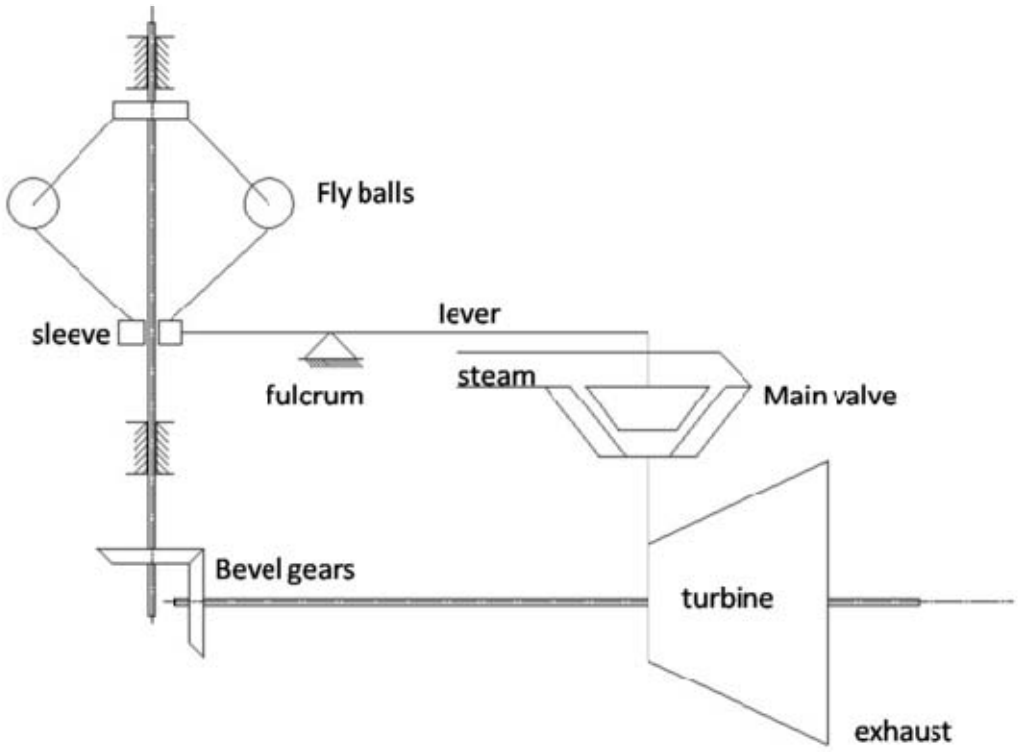


Figure 2.14 Speed governing system

When the generator electrical load is suddenly decreased, the electrical power is then less than the mechanical power input. This power difference causes the turbine speed change and, consequently, the generator frequency to increase. The change in speed is sensed by the turbine governor which acts to adjust the turbine input valve to change the mechanical power input to bring the speed to a new steady-state. Figure 2.15 shows the graphs of turbine torque, electrical torque and speed when the load on a generator suddenly falls.

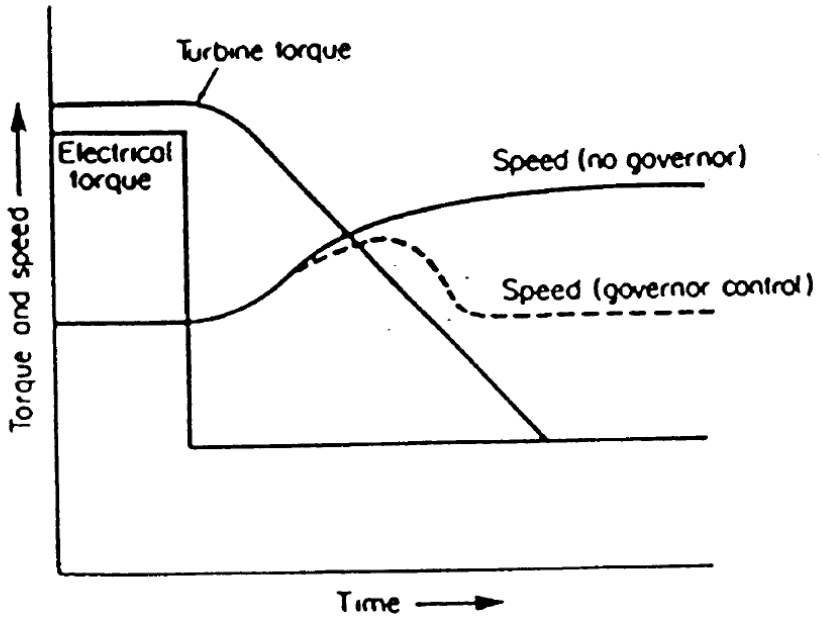


Figure 2.15 Graphs of turbine torque, electrical torque and speed when the load on a generator suddenly falls

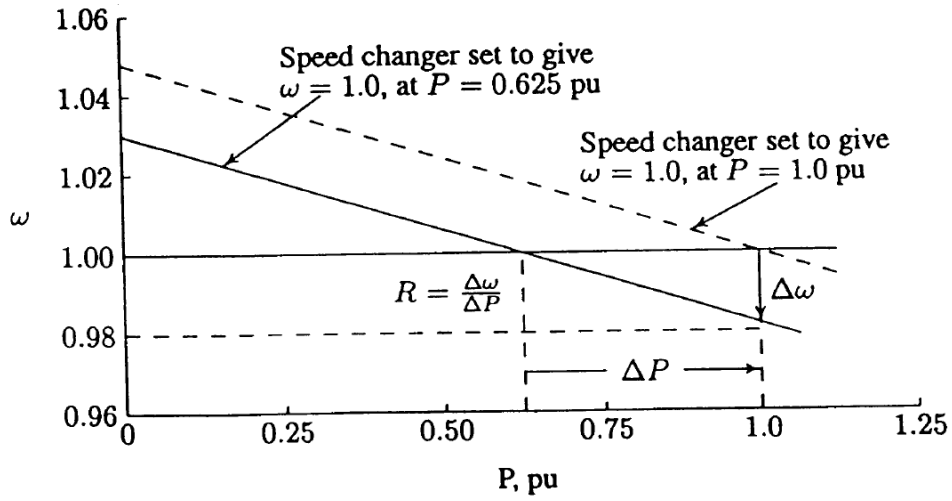


Figure 2.16 Governor steady-state speed characteristics

For a stable operation, the governors are designed to permit the speed to drop as the load increase. The steady-state characteristics of such a governor are shown in Figure 2.16.

The slope of the curve represents the speed regulation, or droop, *R*.

$$R\% = \frac{speed \quad or \quad frequency \quad change}{power \quad output \quad change} \times 100\%$$

$$= \frac{\omega_{NL} - \omega_{FL}}{\omega_0} \times 100\%$$

ω_{NL} = steady-state speed at no-load

ω_{FL} = steady-state speed at full-load

ω_0 = nominal or rated speed

Governors typically have a speed regulation of 5 – 6% from zero to full load. The speed governor output ΔP_g is the difference between the reference set power ΔP_{ref} and the power $\frac{1}{R}\Delta\omega$ and can be described by the equation

$$\Delta P_g = \Delta P_{ref} - \frac{1}{R}\Delta\omega \tag{2.6}$$

or in s-domain

$$\Delta P_g(s) = \Delta P_{ref}(s) - \frac{1}{R}\Delta\Omega(s) \tag{2.7}$$

The s-domain relationship between the steam valve position ΔP_v and the speed governor output ΔP_g is

$$\Delta P_v(s) = \frac{1}{1+\tau_g s}\Delta P_g(s) \tag{2.8}$$

where τ_g is the speed governor time constant.

Figure 2.17 shows the block diagram of the speed governor system for steam turbine.

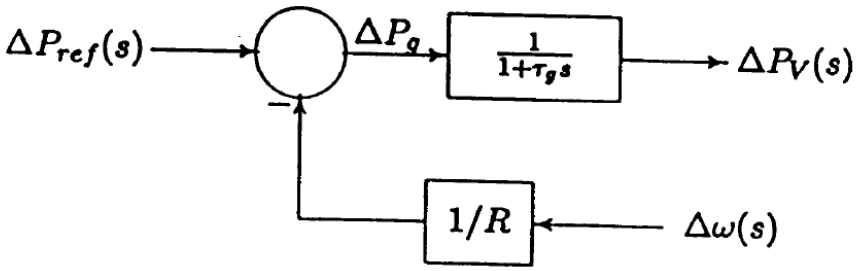


Figure 2.17 Block diagram of the speed governor system for steam turbine

Figure 2.18 shows the block diagram for a single area load frequency control (LFC).

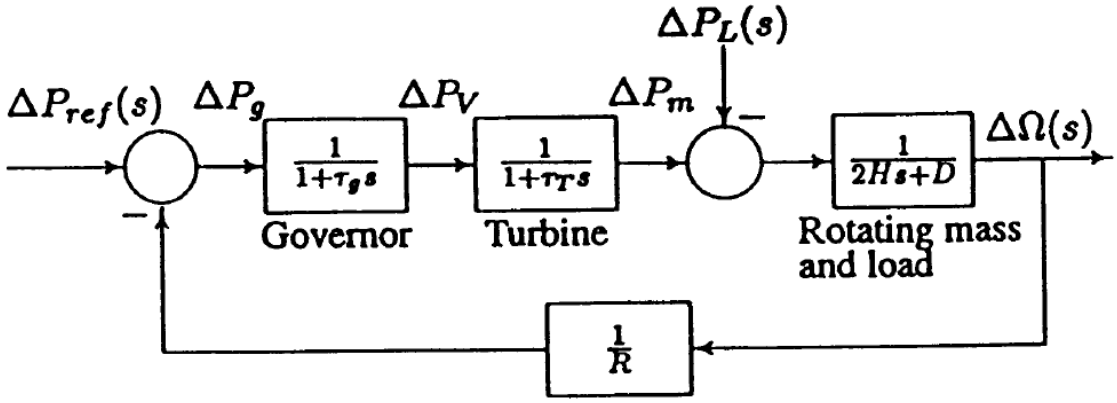


Figure 2.18 Block diagram for a single area LFC.

The transfer function of the system is

$$\frac{\Delta\Omega(s)}{\Delta P_{ref}(s)} = \frac{1}{(2Hs+D)(1+\tau_g s)(1+\tau_T s)+1/R} \tag{2.9}$$

The transfer function relating the load change ΔP_L to the frequency deviation $\Delta\Omega$ is

$$\frac{\Delta\Omega(s)}{\Delta P_L(s)} = -\frac{(1+\tau_g s)(1+\tau_T s)}{(2Hs+D)(1+\tau_g s)(1+\tau_T s)+1/R} \tag{2.10}$$

The load change can be considered as a step input i.e. $\Delta P_L(s) = \frac{\Delta P_L}{s}$. Utilizing the final value theorem, the steady-state value of $\Delta\omega$ is

$$\Delta\omega_{ss} = \lim_{s\to 0} s\Delta\Omega(s) = -\Delta P_L \frac{1}{D+1/R} \tag{2.11}$$

It is clear that for the case with no frequency-sensitive load (i.e. with $D = 0$), the steady-state deviation in frequency is determined by the governor speed regulation constant R and is

$$\Delta\omega_{ss} = -\Delta P_L R$$

When several generators with governor speed regulations, R_1, R_2, …, R_n, and frequency-sensitive load change parameters, D_1, D_2, …, D_n, are connected to the system, the steady-state deviation in frequency is given by

$$\Delta\omega_{ss} = -\Delta P_L \frac{1}{D_1 + D_2 + \cdots + D_n + 1/R_1 + 1/R_2 + \cdots + 1/R_n} \tag{2.12}$$

where defining

$$\beta = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

is called the area frequency response characteristic. The equation (2.12) becomes

$$\Delta\omega_{ss} = -\Delta P_L \frac{1}{D_1 + D_2 + \cdots + D_n + \beta} \tag{2.12a}$$

Example 2.1:

A single area power system, as shown in Figure 2.18, has the following parameters:

$\tau_T = 0.5$ sec.; $\tau_g = 0.2$ sec. and $H = 5$ sec.

The load varies by 0.72 percent for a 0.9 percent change in frequency. Find

(a) What is the value of D?

(b) Use the Roth-Hurwitz array to find the range of R for the system stable; and

(c) If $R = 0.05$ per unit and the turbine rated output is 250 MW at the nominal frequency of 50 Hz, find the system steady-state frequency deviation in Hz when a sudden load increase of 50 MW occurs.

Solution:

(a) $$D = \frac{0.72}{0.9} = 0.8$$

(b) Using the equation (2.9) gives that the characteristic equation is

$$(2Hs + D)(1 + \tau_g s)(1 + \tau_T s) + 1/R = 0$$

Setting $K = 1/R$ and substituting the parameters into the equation gives that

$$s^3 + 7.08s^2 + 10.56s + 0.8 + K = 0$$

The Routh-Hurwitz array for the polynomial is then

s^3	1	10.56
s^2	7.08	0.8+K
s^1	$\frac{73.965-K}{7.08}$	0
s^0	0.8+K	0

From the s^1 row, we can see that for the system stability, K must be less than 73.965. Also from s^0 row, K must be greater than -0.8. Thus, with positive values of K

$$K < 73.965$$

Since $R = 1/K$, for the system stability, the governor speed regulation must be

$$R > \frac{1}{73.965} = 0.0135$$

(c) Using (2.11) gives that

$$\Delta\omega_{ss} = \lim_{s\to 0} s\Delta\Omega(s) = -\Delta P_L \frac{1}{D+1/R}$$

$$= (-\frac{50}{250})\frac{1}{0.8+1/0.05} = (-0.2)\frac{1}{20.8}$$

$$= -0.0096 \quad per\ \ unit$$

Thus, the steady-state frequency deviation in Hz due to the sudden load change of 50 MW is

$$\Delta f = -(0.0096)(50) = -0.48 \qquad Hz$$

Example 2.2:

One area of an interconnected 50 Hz power system has three turbine generator units rated 1000, 750 and 500 MVA, respectively. The regulation constant of each unit is R = 0.05 per unit based on its own rating. Each unit is initially operating at one-half of its own rating, when the system load suddenly increases by 200 MW. Determine

(a) The per unit area frequency response β on a 1000 MVA system base;
(b) The steady-state drop in area frequency; and
(c) The increase in the turbine mechanical power input of each unit.

Assume that the reference power setting of each turbine-generator remains constant. Neglect losses and the dependence of load on frequency.

Solution:

(a) The regulation constant must be converted to the system per unit base using

$$R_{new} = R_{old} \frac{S_{base(new)}}{S_{base(old)}}$$

It gives that

$$R_{1,new} = R_{1,old} = 0.05 \qquad per \quad unit$$

$$R_{2,new} = (0.05)\frac{1000}{750} = 0.06667 \qquad per \quad unit$$

$$R_{3,new} = (0.05)\frac{1000}{500} = 0.10 \qquad per \quad unit$$

$$\beta = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = 45$$

(b) Neglecting losses and the dependence of load on frequency, the steady-state increase in total turbine mechanical power equals the load increase 200 MW ($\Delta P_m = 0.20$ per unit). With $\Delta P_{ref} = 0$

$$\Delta f = -\frac{1}{\beta}\Delta P_m = -\frac{1}{45}(0.2) = -4.444 \times 10^{-3} \qquad per \quad unit$$

$$= (-4.444 \times 10^{-3})(50) = -0.2222 \quad Hz$$

The steady-state frequency drop is 0.2222 Hz.

(c) Using $\Delta f = -4.444 \times 10^{-3}$ per unit

$$\Delta P_{m1} = -\frac{1}{R_1}\Delta f = -\frac{1}{0.05}(-4.4444 \times 10^{-3})$$

$$= 0.08888 \quad per \quad unit = 88.88 \quad MW$$

$$\Delta P_{m2} = -\frac{1}{R_2}\Delta f = -\frac{1}{0.06667}(-4.4444 \times 10^{-3})$$

$$= 0.06667 \quad per \quad unit = 66.67 \quad MW$$

$$\Delta P_{m3} = -\frac{1}{R_3}\Delta f = -\frac{1}{0.10}(-4.4444 \times 10^{-3})$$

$$= 0.04444 \quad per \quad unit = 44.44 \quad MW$$

Note that Unit 1, whose MVA rating is 33.33% larger than that of Unit 2 and 100% larger than Unit 3, pick up 33.33% more load that Unit 2 and 100% more load than Unit 3. That is, each unit shares the total load change in proportion to its own rating.

The main objectives of LFC are to follow a load change and to maintain the steady-state frequency error Δf to be zero. In order to eliminate the steady-state frequency error, we introduce the concept of *Area Control Error* (*ACE*), which is defined as

$$ACE = (P_{tie} - P_{tie,sched}) + B\Delta f = \Delta P_{tie} + B\Delta f \tag{2.13}$$

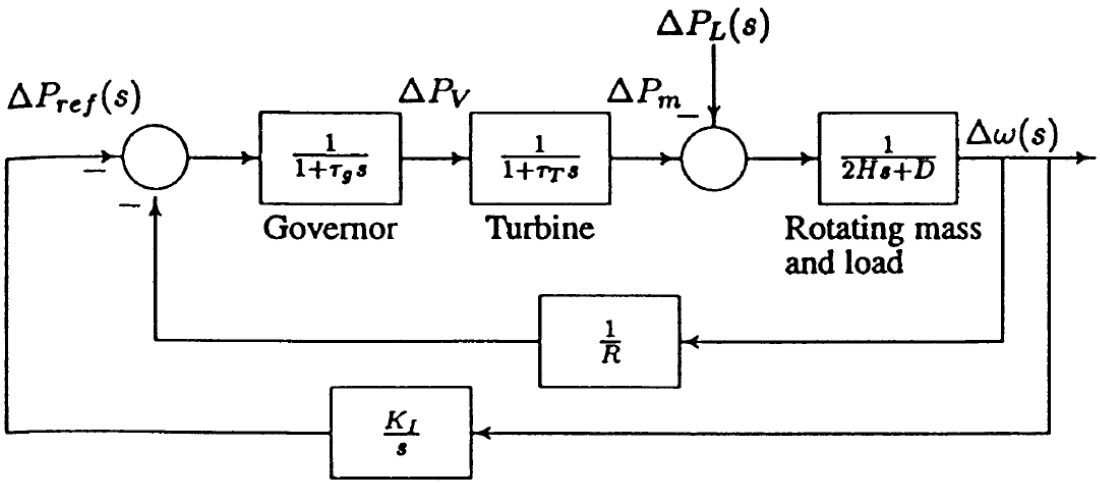
where ΔP_{tie} is the deviation in net tie-line power flow out of the area from its scheduled value $P_{tie,sched}$; Δf is the deviation of the area frequency from 50 Hz and *B* is the frequency bias constant, which determines the amount of interaction during a disturbance in the neighbouring areas.

The change in each reference power setting $\Delta P_{i,ref}$ of each turbine generator operating under LFC is

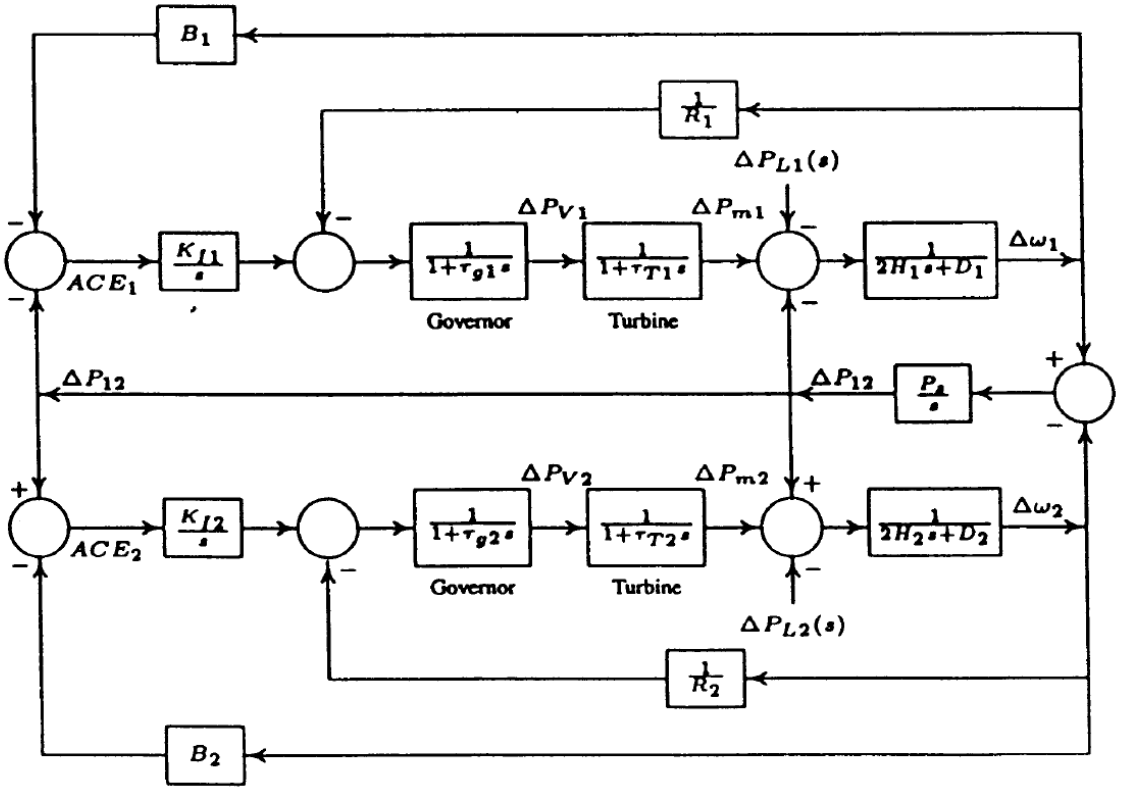
$$\Delta P_{i,ref} = -K_{Ii} \int ACE dt \tag{2.14}$$

The negative sign indicates that if the ACE is negative (either the net tie-line power flow out of the area or the area frequency is low), then the area should increases its generation (raise request).

LFC system block diagram for single area and two-area systems are shown in Figures 2.19 and 2.20, respectively.



Figures 2.19 LFC system block diagram for single area system



Figures 2.20 LFC system block diagram for a two-area system

Consider the LFC single area system, as shown in Figure 2.19, we have

$$\frac{\Delta\Omega(s)}{\Delta P_L(s)} = -\frac{s(1+\tau_g s)(1+\tau_T s)}{s(2Hs+D)(1+\tau_g s)(1+\tau_T s)+K_I+s/R} \quad (2.15)$$

If the load change is a step input, i.e. $\Delta P_L(s) = \frac{\Delta P_L}{s}$, we can obtain that by utilizing the final value theorem, the steady-state value of $\Delta\omega$ is

$$\begin{aligned}\Delta\omega_{ss} &= \lim_{s\to 0} s\Delta\Omega(s) \\ &= -s\frac{\Delta P_L}{s}\frac{s(1+\tau_g s)(1+\tau_T s)}{s(2Hs+D)(1+\tau_g s)(1+\tau_T s)+K_I+s/R} = 0 \quad (2.16)\end{aligned}$$

Introducing the ACE into LFC can maintain the steady-state frequency error Δf to be zero. The ACE is zero in every area only when both ΔP_{tie} and Δf are zero.

Example 2.3:

A two-area system connected by a tie-line has the following parameters on a 1000 MVA common base.

Area	1	2
R	$R_1 = 0.05$	$R_2 = 0.0625$
D	$D_1 = 0.6$	$D_2 = 0.9$
Base Power	1000 MVA	1000 MVA

The units are operating in parallel at the nominal frequency of 50 Hz. A load increase of 187.5 MW occurs in Area 1. Determine the new steady-state frequency and the change in tie-line flow.

Solution:

The per unit load increase in Area 1 is

$$\Delta P_{L1} = \frac{\Delta P_L}{S_{base}} = \frac{187.5}{1000} = 0.1875 \quad per \quad unit$$

The per unit steady-state frequency deviation is

$$\Delta\omega_{ss} = \frac{-\Delta P_L}{(D_1 + 1/R_1) + (D_2 + 1/R_2)}$$

$$= \frac{-0.1875}{(20 + 0.6) + (16 + 0.9)} = -0.005 \quad per \quad unit$$

Thus, the steady-state frequency deviation in Hz is

$$\Delta f = -(0.005)(50) = -0.25 \qquad Hz$$

and the new frequency is

$$f = f_0 + \Delta f = 50 - 0.25 = 49.75 \qquad Hz$$

The change in the mechanical power in each area is

$$\Delta P_{m1} = -\frac{\Delta\omega}{R_1} = -\frac{-0.005}{0.05} = 0.10 \quad per \quad unit = 100 \quad MW$$

$$\Delta P_{m2} = -\frac{\Delta\omega}{R_2} = -\frac{-0.005}{0.0625} = 0.08 \quad per \quad unit = 80 \quad MW$$

Thus, Area 1 increases the generation by 100 MW and Area 2 by 80 MW at the new operating frequency 49.75 Hz. The total change in generation is 180 MW, which is 7.5 MW less than the load change because of the change in the area loads due to the frequency drops.

The change in Area 1 load is

$$\Delta\omega D_1 = -(0.005)(0.6) = -0.003 \quad per \quad unit = -3.0 \quad MW$$

and the change in Area 2 is

$$\Delta\omega D_2 = -(0.005)(0.9) = -0.0045 \quad per \quad unit = -4.5 \quad MW$$

Thus, the change in the total area load is -7.5 MW.

The tie-line power follow is

$$\Delta P_{12} = \Delta\omega(\frac{1}{R_2} + D_2) = (-0.005)(16.9)$$
$$= -0.0845 \quad per \quad unit = -84.5 \quad MW$$

That is, 84.5 MW flows from Area 2 to Area 1. 80 MW comes from the increased generation in Area 2, and 4.5 MW comes from the load reduction in Area 2 due to the frequency drop.

Question:

Can you re-solve Example 2.3 if the system parameters are?

Area	1	2
R	$R_1 = 0.05$	$R_2 = 0.0625$
D	$D_1 = 0.6$	$D_2 = 0.9$
Base Power	1000 MVA	750 MVA

Use 1000 MVA as the common base.

2.3 Economic Dispatch

Economic dispatch (ED) refers to classical approaches to the economic operation of power systems. A key element here is the proper modeling of power plant efficiency.

2.3.1 Heat Rate Curve and Cost Rate Curve

Normally, on models the heat rate input to the boiler as a function of output real power. The measured heat rate curve can be quite complex depending on the valve positions of the steam turbines. The heat rate characteristics is also dependent on thermodynamic parameters like ambient dry and wet bulb temperatures, operating pressures, water pumping rates, etc. This applies not only to the characteristic curve itself but also to upper and lower limits on generation. Power utilities normally utilize a single heat rate curve with upper and lower limits on generation. The curves is approximated, usually, by a quadratic polynomial of the form

$$HR(P_G) = a + bP_G + cP_G^2 \tag{2.17}$$

where P_G is the MW (or per unit) output power of the generator and a, b and c are constant coefficients. Where there are wide seasonal variations in ambient conditions, more than one heat rate characteristic may be necessary. This will apply also to lower and upper limits on generation.

The heat rate characteristic is useful in describing thermodynamic efficiency. In economic operation, however, one is interested in the cost fuel needed to produce the required power. By knowing the type of fuel used, together

with its calorific value and cost, one can translates the heat rate curve to a cost rate one. Figure 2.21 and Figure 2.22 show the heat rate curve and cost rate curve, respectively.

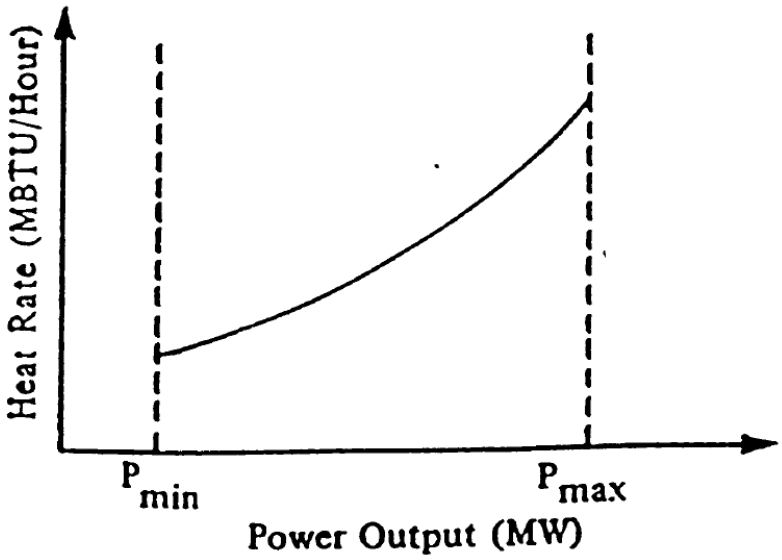


Figure 2.21 Heat rate curve

Multiply Fuel cost ($/MBTU)

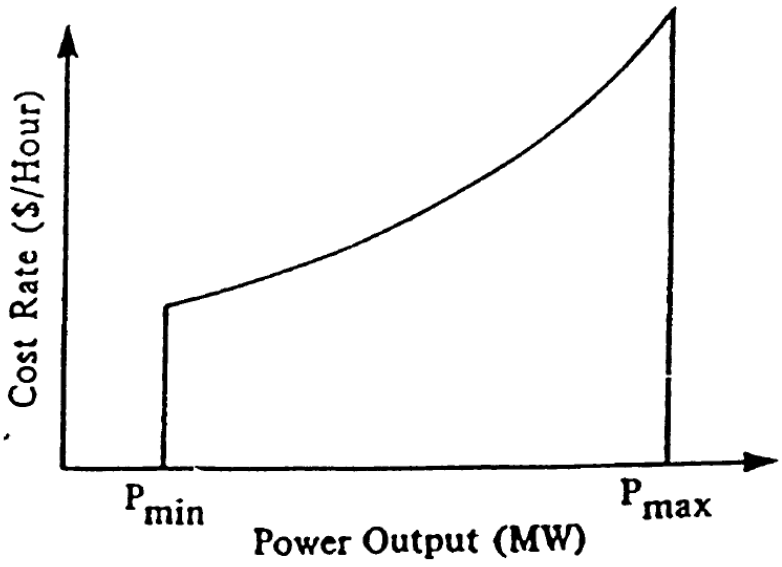


Figure 2.22 Cost rate curve

The cost rate curve is also approximated by a polynomial of the form

$$CR(P_G) = \bar{a} + \bar{b}P_G + \bar{c}P_G^2 \tag{2.18}$$

where P_G is the MW (or per unit) output power of the generator and $\bar{a}$, $\bar{b}$ and $\bar{c}$ are constant coefficients.

We are also concerned with the incremental cost for the heat rate and the cost rate since they are determined by the slopes of the curves shown in Figure 2.21 and Figure 2.23. If the heat rate (HR) curve and the cost rate (CR) curve can be represented by equations (2.17) and (2.18), respectively, the incremental heat rate (*IHR*) and the incremental cost rate (*ICR*) will be

$$IHR(P_G) = \frac{dHR}{dP_G} = b + 2cP_G \tag{2.19}$$

and

$$ICR(P_G) = \frac{dCR}{dP_G} = \bar{b} + 2\bar{c}P_G \tag{2.20}$$

In the utilities, instead of polynomial curves, piecewise linear heat rate and cost rate curves may be used, as shown in Figure 2.23.

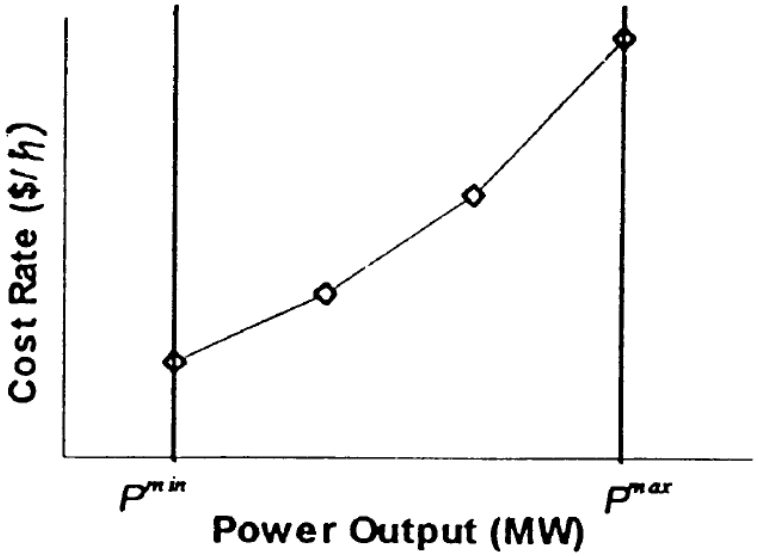


Figure 2.23 Piecewise linear cost rate curve

Piecewise incremental heat rate (IHR) curve or incremental cost rate (ICR) curve is shown in Figure 2.24.

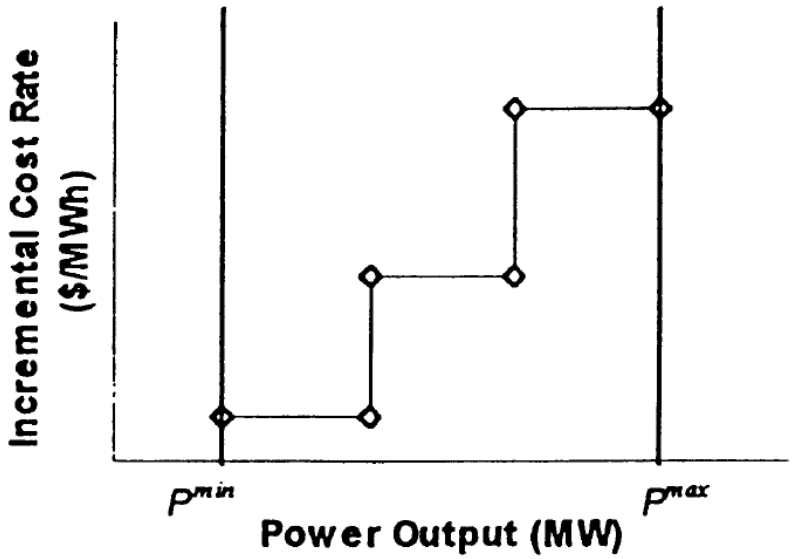


Figure 2.24 Piecewise linear incremental cost rate curve

Example 2.4:

Given:

$$HR(P_G) = 510 + 7.2P_G + 0.00142P_G^2$$

$$IHR(P_G) = 7.2 + 0.00284P_G$$

and the fuel cost = \$2/*MBTU*.
Find $CR(P_G)$ and $ICR(P_G)$.

Solution:

We have

$$CR(P_G) = fuel \quad \cos t \times HR(P_G)$$

and

$$ICR(P_G) = fuel \quad \cos t \times IHR(P_G)$$

It follows that

$$CR(P_G) = 1020 + 14.4P_G + 0.00284P_G^2$$

$$ICR(P_G) = 14.4 + 0.00568P_G$$

There are two important cases of optimal economic dispatch:

1. Economic dispatch neglecting losses. The transmission losses can be neglected.
2. Economic dispatch including transmission losses. Large electric power systems have many widely separated generating units which are interconnected by long transmission lines, it is necessary to consider the losses in determining system optimal economic dispatch.

2.3.2 Lossless Thermal Dispatch

Consider a power plant with n generating units, if transmission losses are neglected, then the total real power demand P_D is related to generation by the simple equality

$$\sum_{i=1}^{n} P_{Gi} = P_D \tag{2.21}$$

The cost associated with the total generation is given by

$$CR = CR_1(P_{G1}) + CR_2(P_{G2}) + \cdots + CR_n(P_{Gn}) \tag{2.22}$$

If one assumes that the generation limits are not violated, the necessary condition of optimality is obtained as follows: Determine the LaGrangian

$$L = \sum_{i=1}^{n} CR_i(P_{Gi}) + \lambda(P_D - \sum_{i=1}^{n} P_{Gi}) \tag{2.23}$$

and then set its derivatives with respect to P_{Gi} and λ to zero, i.e.

$$0 = \frac{\partial L}{\partial P_{Gi}} = \frac{dCR_i}{dP_{Gi}} - \lambda \tag{2.24}$$

$$0 = \frac{\partial L}{\partial \lambda} = P_D - \sum_{i=1}^{n} P_{Gi} \tag{2.25}$$

for $i = 1, 2, \ldots, n$.

The derivatives $\frac{dCR_i}{dP_{Gi}}$ (i = 1, 2, …, n) are known as the incremental costs of the ith generation unit. From a physical point of view, the incremental cost represents the cost (in \$/*MWH*) of generating the next *MWH* at the generation level of P_{Gi}. From the above necessary optimality conditions, it is clear that at the optimum levels of generation on has

$$\lambda = \frac{dCR_i}{dP_{Gi}} \tag{2.26}$$

for all i = 1, 2, …, n. It states that *all generating units operate at equal incremental cost.*

Example 2.5:

In a system where the transmission losses are neglected, the load of 10.0 pu is supplied from two generators *G1* and *G2*. The costs of generation are given by

$$CR_1(P_{G1}) = 0.5 + P_{G1} + P_{G1}^2$$

$$CR_2(P_{G2}) = 1.5 + 0.5P_{G2} + 2P_{G2}^2$$

Determine the optimum values of P_{G1} and P_{G2} such that the demand is met.

Solution:

The requirement that the demand is met is given by

$$P_{G1} + P_{G2} = 10.0$$

The cost function is

$$\begin{aligned} CR &= CR_1(P_{G1}) + CR_2(P_{G2}) \\ &= 2.0 + P_{G1} + P_{G1}^2 + 0.5P_{G2} + 2P_{G2}^2 \end{aligned}$$

Solution 1: This is an optimum problem with equality constraints only. The LaGrangian is expressed as follows

$$\begin{aligned} L &= CR_1(P_{G1}) + CR_2(P_{G2}) + \lambda(P_D - P_{G1} - P_{G2}) \\ &= 2.0 + P_{G1} + P_{G1}^2 + 0.5P_{G2} + 2P_{G2}^2 + \lambda(10.0 - P_{G1} - P_{G2}) \end{aligned}$$

The necessary optimality conditions require that

$$0 = \frac{\partial L}{\partial P_{G1}} = \frac{dCR_1}{dP_{G1}} - \lambda = 1 + 2P_{G1} - \lambda$$

$$0 = \frac{\partial L}{\partial P_{G2}} = \frac{dCR_2}{dP_{G2}} - \lambda = 0.5 + 4P_{G2} - \lambda$$

$$0 = \frac{\partial L}{\partial \lambda} = P_D - P_{G1} - P_{G2} = 10.0 - P_{G1} - P_{G2}$$

The solution is obtained by expressing P_{G1} and P_{G2} in terms of λ

$$P_{G1} = 0.5\lambda - 0.5$$

$$P_{G2} = 0.25\lambda - 0.125$$

Substituting them into the previous equations gives that

$$\lambda = 14.167 \ \$/MWH$$

It follows that

$$P_{G1} = 6.5833 \quad per \quad unit$$

$$P_{G2} = 3.4167 \quad per \quad unit$$

Solution 2: Using the conclusion that that *all generating units operate at equal incremental cost*.

$$ICR_1 = ICR_2$$

It gives that

$$ICR_1(P_{G1}) = 1 + 2P_{G1} = 0.5 + 4P_{G2} = ICR_2(P_{G2})$$

It follows that

$$2P_{G1} - 4P_{G2} = -0.5$$

We have that $P_{G1} + P_{G2} = 10.0$.

It follows that

$$P_{G1} = 6.5833 \quad per \quad unit$$

$$P_{G2} = 3.4167 \quad per \quad unit$$

and $\lambda = 14.167$ \$/*MWH*

Economic Dispatch Including Generator Limits

In the real world, generators have limits on their generation, both minimum and maximum generation. Thus we have the inequality conditions:

$$P_{i(\min)} \le P_i \le P_{i(\max)} \qquad i = 1,\ 2,\ .....,\ n$$

which are in addition to the equality condition (2.21)

$$\sum_{i=1}^{n} P_i = P_D$$

The necessary conditions for optimal dispatch in lossless cases but including generator limits become:

$$\frac{dCR_i}{dP_i} = \lambda \quad for \quad P_{i(\min)} \le P_i \le P_{i(\max)}$$

$$\frac{dCR_i}{dP_i} \le \lambda \quad for \quad P_i = P_{i(\max)}$$

$$\frac{dCR_i}{dP_i} \ge \lambda \quad for \quad P_i = P_{i(\min)}$$

If the calculated production $P_i > P_{i\,max}$, then set $P_i = P_{i\,max}$
If the calculated production $P_i < P_{i\,min}$, then set $P_i = P_{i\,min}$

Example 2.6:

Incremental costs in \$/*MWH* for a power plant consisting of two units are given by

$ICR_1 = 8.0 + 0.0080\,P_{G1}$ and $ICR_2 = 6.4 + 0.0096\,P_{G2}$

Assume that both units are operating at all times, that total load varies from 250 MW to 1250 MW, and that maximum and minimum loads on each unit are to be 625 MW and 100 MW, respectively. Find the incremental cost and the allocation of load between units for the minimum cost of various total loads.

Solution:

At light loads Unit 1 will have the higher incremental cost and will operate at its lower limit of 100 MW for which the ICR_1 is \$8.8 per MWH. When the output of Unit 2 is also 100 MW, ICR_2 is \$7.36 per MWH. There, as the plant output increases, the additional output should come from Unit 2 until the ICR_2 becomes \$8.8 per MWH. Until that point is reached, the incremental cost λ of the plant is determined by Unit 2 alone. When the plant load is 250 MW, Unit 2 will supply 150 MW with the ICR_2 is \$7.84 per MWH. When the ICR_2 is \$8.8 per MWH ($ICR_2 = 6.4 + 0.0096\,P_{G2} = 8.8$), it gives that

$$P_{G2} = \frac{8.8 - 6.4}{0.0096} = 250 \quad MW$$

and total plant output is 350 MW. From this point on the required out of each unit for economic load distribution is found by assuming various values of λ and calculating each

unit's output and the total plant output. The results are shown in Table 2.1.

Table 2.1 Output of each unit and λ values

λ \$/MWH	P_{G1} MW	P_{G2} MW	$P_{G1} + P_{G2}$ MW
8.8; 7.84	100	150	250
8.8*	100	250	350
9.6	200	333	533
10.4	300	417	717
11.2	400	500	900
12.0	500	583	1083
12.4*	550	625	1175
13.0, 12.4	625	625	1250

Table 2.1 shows that λ = 12.4 \$/*MWH*, Unit 2 is operating at its upper limit, and that additional load must come from Unit 1 alone, which then determines the values of λ.

If we want to know the distribution of load between the units for a plant output of 1000 MW, we have

$$P_{G1} + P_{G2} = 1000$$

and

$$8.0 + 0.0080P_{G1} = 6.4 + 0.0096(1000 - P_{G1})$$

It gives that

$$P_{G1} = 454.55 \quad MW$$

$$P_{G2} = 545.45 \quad MW$$

and for each unit $\lambda = 11.636$ \$/*MWH*

Example 2.7:

Determine the saving in cost in \$/*MWH* for the economic distribution of a total load of 900 MW between the two units of the plant described in Example 2.6, compared with the equal distribution of the same total load.

Solution:

Example 2.6 shows that Unit 1 should supply 400 MW and Unit 2 should supply 500 MW. If each unit supplies 450 MW, the increase in cost for Unit should be

$$\int_{400}^{450}(8.0+0.0080P_{G1})dP_{G1}=(8.0P_{G1}+0.004P_{G1}^2)\Big|_{400}^{450}$$

$$=\$570 \quad per \quad hour$$

Similarly, Unit 2

$$\int_{500}^{450}(6.4+0.0096P_{G2})dP_{G2}=(6.4P_{G2}+0.0048P_{G2}^2)\Big|_{500}^{450}$$

$$=-\$548 \quad per \quad hour$$

The negative sign means a decrease in cost, as we expect for a decrease in output form Unit 2. The net increase in cost for the plant is

\$570 - \$548 = \$22 *per hour*

The saving seems small, but this amount saved every hour for a year of continuous operation would reduce cost by \$192,720 for the year.

General expression:

For a plant with *n* units, considering (2.18), (2.23) and (2.26) gives that

$$CR_i(P_{Gi}) = \overline{a}_i + \overline{b}_i P_{Gi} + \overline{c}_i P_{Gi}^2$$

$$L = \sum_{i=1}^{n} CR_i(P_{Gi}) + \lambda(P_D - \sum_{i=1}^{n} P_{Gi})$$

and

$$\lambda = \frac{dCR_i}{dP_{Gi}}$$

for all $i = 1, 2, \ldots, n$.

Solving for P_{Gi} in terms of λ gives that

$$\lambda = \frac{dCR_i}{dP_{Gi}} = \overline{b}_i + 2\overline{c}_i P_{Gi} \tag{2.27}$$

Hence

$$P_{Gi} = \frac{\lambda - \overline{b}_i}{2\overline{c}_i} \tag{2.28}$$

Using the original equality constraint gives that

$$P_D = \sum_{i=1}^{n} P_{Gi} = \sum_{i=1}^{n} \frac{\lambda - \overline{b}_i}{2\overline{c}_i} = \alpha\lambda - \beta \tag{2.29}$$

where $\alpha = \sum_{i=1}^{n} \frac{1}{2\overline{c}_i}$ $\beta = \sum_{i=1}^{n} \frac{\overline{b}_i}{2\overline{c}_i}$

Thus,

$$\lambda = \frac{1}{\alpha}(P_D + \beta) \tag{2.30}$$

As a result

$$P_{Gi} = \frac{1}{2\bar{c}_i}[\frac{1}{\alpha}(P_D + \beta) - \bar{b}_i] \tag{2.31}$$

Thus, in the simple lossless case, the optimal levels of generation can be computed in a closed-form fashion. At those levels they all have equal incremental costs.

Example 2.8: Calculate the generations of the two units in Example 2.6 for a plant output of 1000 MW

Solution: We have

$$P_{G1} + P_{G2} = 1000$$

$$\bar{b}_1 = 8.0 \qquad and \qquad 2\bar{c}_1 = 0.008$$

$$\bar{b}_2 = 6.4 \qquad and \qquad 2\bar{c}_2 = 0.0096$$

$$\frac{1}{2\bar{c}_1} = 125, \qquad \frac{1}{2\bar{c}_2} = 104.167$$

$$\alpha = \sum_{i=1}^{2} \frac{1}{2\bar{c}_i} = 229.167, \qquad \beta = \sum_{i=1}^{2} \frac{\bar{b}_i}{2\bar{c}_i} = 1666.667$$

$$\lambda = \frac{1}{\alpha}(P_D + \beta) = \frac{1}{229.167}(1000 + 1666.667) = 11.636$$

$$P_{G1} = \frac{1}{2\bar{c}_1}(\lambda - \bar{b}_1) = 125 \times (11.636 - 8.0) = 454.55 \quad MW$$

$$P_{G2} = 1000 - P_{G1} = 545.45 \quad MW$$

2.3.3 Thermal Dispatch with Losses

Since transmission losses cannot be ignored, early workers have attempted to include the effect of losses in an approximate manner. Normally, losses depend on the transmission network, the distribution of loads and generation levels. Symbolically one can express the transmission losses as a function of system generation, i.e.

$$P_L = f(P_{G1},\ P_{G2},\ \cdots,\ P_{Gn})$$

As a result, total generation equals to the load P_D plus losses P_L, i.e.

$$\sum_{i=1}^{n} P_{Gi} = P_D + P_L$$

In deriving the necessary optimality condition for this problem, one can writes the LaGrangian as

$$L = \sum_{i=1}^{n} CR_i(P_{Gi}) + \lambda(P_D + P_L - \sum_{i=1}^{n} P_{Gi}) \tag{2.32}$$

The necessary conditions become

$$0 = \frac{\partial L}{\partial P_{Gi}} = \frac{dCR_i}{dP_{Gi}} - \lambda(1 - \frac{\partial P_L}{\partial P_{Gi}}) \tag{2.33}$$

$$0 = \frac{\partial L}{\partial \lambda} = P_D + P_L - \sum_{i=1}^{n} P_{Gi} \tag{2.34}$$

for all i = 1, 2, …, n.

From these necessary conditions, one can obtains

$$\lambda = \frac{1}{1 - \dfrac{\partial P_L}{\partial P_{Gi}}} \frac{dCR_i}{dP_{Gi}} \tag{2.35}$$

The quantities

$$L_i = \frac{1}{1 - \dfrac{\partial P_L}{\partial P_{Gi}}}$$

are referred to as the *penalty factors*. Since the dependence of P_L on P_{G1}, P_{G2}, …, P_{Gn} is not simple to establish, the solution to this problem cannot be obtained in a closed form.

(2.35) can be rewritten as

$$\lambda = L_i \frac{dCR_i}{dP_{Gi}} = L_i \times ICR_i \tag{2.36}$$

Thus the optimum operation is achieved when the product of the incremental cost times the penalty factor is the same for all plants.

Since $L_i \geq 1$ ($1 > \dfrac{\partial P_L}{\partial P_{Gi}} \geq 0$), it gives that

- $L_i \approx 1$: generator *i* is closed to the loads and most of the increased generation goes to supply the load demand.
- $L_i > 1$: generator *i* is far away from loads and is required to supply an increased load demand in the presence of high transmission losses.

Example 2.9:

A three-unit system has the following cost rate curves:

$$CR_1 = 323.634 + 7.424P_{G1} + 0.0035P_{G1}^2$$
$$CR_2 = 565.230 + 6.972P_{G2} + 0.0028P_{G2}^2$$
$$CR_3 = 602.105 + 6.717P_{G3} + 0.0030P_{G3}^2$$

(a) Determine the economic operating points for these three units when delivering a total load of 745 MW.

(b) Repeat the Part (a) with using the transmission losses

$$P_L = 0.00003P_{G1}^2 + 0.00009P_{G2}^2 + 0.00012P_{G3}^2$$

Solution:

(a) We have

$$ICR_1 = 7.424 + 0.0070P_{G1}$$
$$ICR_2 = 6.972 + 0.0056P_{G2}$$
$$ICR_3 = 6.717 + 0.0060P_{G3}$$

It follows that

$$ICR_1 = ICR_2 = ICR_3 = \lambda$$

It gives that

λ = 8.5435 \$/MWh
P_{G1} = 159.938 MW
P_{G2} = 280.635 MW
P_{G3} = 304.427 MW
$P_{G1} + P_{G2} + P_{G3}$ = 745 MW

(b) From the transmission losses

$$\frac{\partial P_L}{\partial P_{G1}} = 2 \times 0.00003 P_{G1}$$
$$\frac{\partial P_L}{\partial P_{G2}} = 2 \times 0.00009 P_{G2}$$
$$\frac{\partial P_L}{\partial P_{G3}} = 2 \times 0.00012 P_{G3}$$

and $P_{G1} + P_{G2} + P_{G3} = 745 + P_L$

$$\lambda(1 - \frac{\partial P_L}{\partial P_{Gi}}) = ICR_i$$

It follows that

$$7.424 + 0.0070P_{G1} = \lambda\,[1 - 2(0.00003)\,P_{G1}]$$
$$6.972 + 0.0056P_{G2} = \lambda\,[1 - 2(0.00009)\,P_{G2}]$$
$$6.717 + 0.0060P_{G3} = \lambda\,[1 - 2(0.00012)\,P_{G3}]$$

The selected initial values are
P_{G1} = 150 MW, P_{G2} = 300 MW and P_{G3} = 295 MW
P_L = 19.218 MW

It gives that

$$7.424 + 0.0070P_{G1} = 0.9910\,\lambda$$
$$6.972 + 0.0056P_{G2} = 0.9460\,\lambda$$
$$6.717 + 0.0060P_{G3} = 0.9292\,\lambda$$

$P_{G1} + P_{G2} + P_{G3} = 745 + 19.218 = 764.218$
These equations are linear equations, so they can be solved for λ directly. It gives that

P_{G1} = 213.87 MW, P_{G2} = 275.71 MW and P_{G3} = 274.62 MW
λ = 9.0021 $/MWh

Table 2.2 shows the iteration results

Table 2.2 Iteration Results

No	P_{G1} (MW)	P_{G2} (MW)	P_{G3} (MW)	P_L (MW)	λ ($/MWh)
1	150.00	300.00	295.00	19.218	9.0021
2	213.87	275.71	274.62	17.263	8.9781
3	205.46	278.55	278.24	17.539	8.9819
4	206.65	278.39	277.48	17.495	8.9811
5	206.54	278.27	277.67	17.500	8.9810

P_{G1} = 206.54 MW, P_{G2} = 278.27 MW and P_{G3} = 277.67 MW
λ = 8.9810 $/MWh

It gives that the penalty factors are
L_1 = 1.0125, L_2 = 1.0527 and L_3 = 1.0714

Among the three machines, L_1 is the smallest and L_3 is the largest so that the generation of #1 machine will be increased and that of #3 machine will be reduced more. Figure 2.25 shows the comparison results.

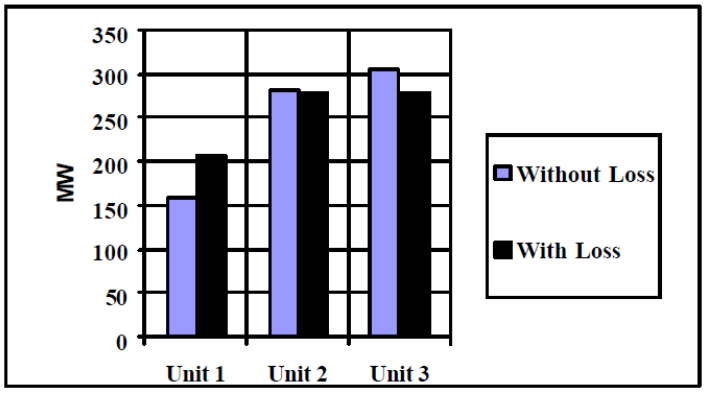


Figure 2.25 Comparison results

Example 2.10:

For a power network shown in Figure 2.26,

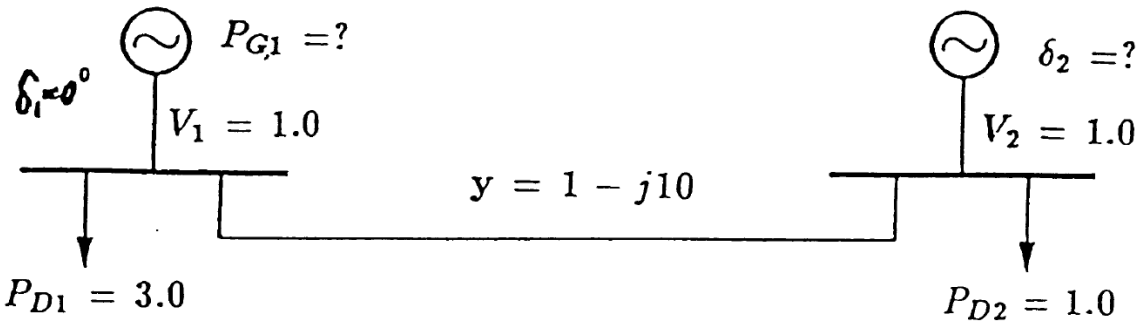


Figure 2.26 Two machine network with loads

$$CR_1(P_{G1}) = 1 + P_{G1} + 3P_{G1}^2$$
$$CR_2(P_{G2}) = 0.5 + 0.5P_{G2} + 0.5P_{G2}^2$$

Determine the optimum levels of generation taking into account system losses.

Solution:

At Bus 1, the real power flow P_{12} is

$$P_{12} = P_{G1} - P_{D1} = P_{G1} - 3.0 = 1 - (\cos\delta_2 + 10\sin\delta_2)$$

and at Bus 2

$$P_{21} = P_{G2} - P_{D2} = P_{G2} - 1.0 = 1 - (\cos\delta_2 - 10\sin\delta_2)$$

Appendix provides the details for the derivation of the power flow equation.

Transmission losses are

$$P_L = P_{12} + P_{21} = 2(1-\cos\delta_2) \qquad \cos\delta = 1 - 2\sin^2\frac{\delta}{2}$$

$$\approx 2(1-(1-\frac{\delta_2^2}{2})) = \delta_2^2 \qquad \approx 1 - 2[\frac{\delta^2}{4}]$$

$$= 1 - \frac{\delta^2}{2}$$

On the other hand,

$$P_{12} - P_{21} = P_{G1} - P_{D1} - P_{G2} + P_{D2}$$

$$= P_{G1} - P_{G2} - 2 = -20\sin\delta_2 \approx -20\delta_2$$

Hence

$$\delta_2 = -\frac{P_{G1} - P_{G2} - 2}{20}$$

Therefore

$$P_L = \delta_2^2 = \frac{1}{400}(P_{G1} - P_{G2} - 2)^2$$

From this relation, one can obtain the partial derivatives of P_L with respect to P_{G1} and P_{G2}

$$\frac{\partial P_L}{\partial P_{G1}} = \frac{1}{200}(P_{G1} - P_{G2} - 2)$$

$$\frac{\partial P_L}{\partial P_{G2}} = -\frac{1}{200}(P_{G1} - P_{G2} - 2)$$

Consequently, the necessary optimality conditions are

$$0 = \frac{\partial L}{\partial P_{G1}} = 1 + 6P_{G1} - \lambda(1 - \frac{1}{200}(P_{G1} - P_{G2} - 2))$$

$$0 = \frac{\partial L}{\partial P_{G2}} = 0.5 + P_{G2} - \lambda(1 + \frac{1}{200}(P_{G1} - P_{G2} - 2))$$

$$0 = \frac{\partial L}{\partial \lambda} = 4 - (P_{G1} + P_{G2} - \frac{1}{400}(P_{G1} - P_{G2} - 2)^2)$$

Define

$$f_1 = 1 - \frac{1}{200}(P_{G1} - P_{G2} - 2)$$

$$f_2 = 1 + \frac{1}{200}(P_{G1} - P_{G2} - 2)$$

Substituting these two questions into $\frac{\partial L}{\partial P_{G1}}$ and $\frac{\partial L}{\partial P_{G2}}$ gives that

$$0 = \frac{\partial L}{\partial P_{G1}} = 1 + 6P_{G1} - \lambda f_1$$

$$0 = \frac{\partial L}{\partial P_{G2}} = 0.5 + P_{G2} - \lambda f_2$$

$$0 = \frac{\partial L}{\partial \lambda} = 4 - (P_{G1} + P_{G2} - \frac{1}{400}(P_{G1} - P_{G2} - 2)^2)$$

These equations are nonlinear. As a result, the solution is obtained by means of an adequate iterative method.

The iterative steps.

Step 1: Initialization

In this step we ignore transmission losses and solve the optimization problem with $P_L = 0$.

The results consist of the following initial guesses for the subsequent iterative steps.

$$P_{G1}^0 = 0.5 \qquad P_{G2}^0 = 3.5 \qquad \lambda^0 = 4$$

Step 2: First Iteration

First, obtain an estimate of losses, given P_{G1}^0 and P_{G2}^0, by means of relation

$$P_L^0 = \frac{1}{400}(P_{G1}^0 - P_{G2}^0 - 2)^2 = 0.0625$$

Then, complete

$$f_1^{\,0} = 1 - \frac{1}{200}(P_{G1}^0 - P_{G2}^0 - 2) = 1.025$$

$$f_2^{\,0} = 1 + \frac{1}{200}(P_{G1}^0 - P_{G2}^0 - 2) = 0.975$$

Based on the above values, the next series of computation follow

$$\lambda^1 = \lambda^0 + \tau \left.\frac{\partial L}{\partial \lambda}\right|_{P_{G1}^0,\ P_{G2}^0} = 4 + \tau(4 + 0.0625 - 4)$$

where τ is an iteration step size factor.

Choosing $\tau = 2$ gives that

$$\lambda^1 = 4.125$$

Consequently,

$$P_{G1}^1 = \frac{1}{6}(\lambda^1 \times f_1^0 - 1) = 0.5381$$
$$P_{G2}^1 = \lambda^1 \times f_2^0 - 0.5 = 3.5219$$

Step 3: Convergence Check

Computing the losses after the first iteration, we get

$$P_L^1 = \frac{1}{400}(P_{G1}^1 - P_{G2}^1 - 2)^2 = 0.062$$

The convergence test involves the error

$$\left|P_{G1}^1 + P_{G2}^1 - P_L^1 - 4\right| = 0.002$$

If this is acceptable, then stop. Otherwise, Step 2 and Step 3 are repeated using the updated values of P_{G1} and P_{G2} until convergence is attained. Table 2.3 shows the iteration results.

Table 2.3 Iteration Results

Iteration	λ	P_{G1}	P_{G2}	P_L	Error $P_D + P_L - P_{G1} - P_{G2}$
1	4.5	0.602	3.888	0.0698	-0.4202
2	4.4	0.586	3.784	0.0675	-0.3025
3	4.14	0.541	3.532	0.0623	-0.0107
4	4.13	0.539	3.527	0.0622	-0.0038
5	4.12				

Define

$$Er = P_D + P_L - P_{G1} - P_{G2}$$

for iteration $k \geq 2$, we have

$$\lambda^{(k+1)} = \Delta\lambda + \lambda^{(k)} = \left|\frac{\lambda^{(k)} - \lambda^{(k-1)}}{Er^{(k)} - Er^{(k-1)}}\right| Er^{(k)} + \lambda^{(k)}$$

Then

$$\lambda^{(3)} = \left|\frac{\lambda^{(2)} - \lambda^{(1)}}{Er^{(2)} - Er^{(1)}}\right| Er^{(2)} + \lambda^{(2)}$$

$$= \left|\frac{4.4 - 4.5}{-0.3025 - (-0.4202)}\right| \times (-0.3025) + 4.4 = 4.14$$

and

$$\lambda^{(4)} = \left|\frac{4.14 - 4.4}{-0.0107 - (-0.3025)}\right| \times (-0.0107) + 4.14 = 4.13$$

Appendix

Derivation of the power flow equations in Example 2.10.

$$P_i = \sum_{k=1}^{2} |V_i||V_k|(G_{ik}\cos\delta_{ik} + B_{ik}\sin\delta_{ik}) = P_{Gi} - P_{Di}$$

where $i = 1, 2$

$$Y_{ik} \triangleq G_{ik} + jB_{ik}$$

$$V_i \triangleq |V_i|e^{j\delta_i} = |V_i|\angle\delta_i$$

$$\theta_{ik} \triangleq \delta_i - \delta_k$$

Recall $e^{j\delta} = \cos\delta + j\sin\delta$ $\qquad \delta_1 = 0$ deg

$Y_{11} = y = 1 - j10$, $Y_{12} = -y = -1 + j10$. $V_1 = V_2 = 1$

$$P_{12} = V_1V_1[G_{11}\cos\delta_{11} + B_{11}\sin\delta_{11}] + V_1V_2[G_{12}\cos\delta_{12} + B_{12}\sin\delta_{12}]$$
$$= 1 + [-\cos(-\delta_2) + 10\sin(-\delta_2)]$$
$$= 1 - (\cos\delta_2 + 10\sin\delta_2)$$

$\delta_{11} = \delta_1 - \delta_1 = 0$ deg

$\delta_{12} = \delta_1 - \delta_2 = -\delta_2$

$$P_{21} = V_2V_2[G_{22}\cos\delta_{22} + B_{22}\sin\delta_{22}] + V_2V_1[G_{21}\cos\delta_{21} + B_{21}\sin\delta_{21}]$$
$$= 1 + (-\cos\delta_2 + 10\sin\delta_2)$$
$$= 1 - (\cos\delta_2 - 10\sin\delta_2)$$

$\delta_{22} = \delta_2 - \delta_2 = 0$ deg

$\delta_{21} = \delta_2 - \delta_1 = \delta_2$

2.4 Coordination of Economic Dispatch with Load Frequency Control

Load frequency control (LFC) with integral controller can achieve zero steady state frequency error and a fast dynamic response, but it exercises no control over the relative loadings of various generating units (i.e. economic dispatch) of the control area. For example, if a sudden small increase in load (say 1%) occurs in the control area, the LFC changes the speed changer settings of the governors of all generating units in the area, so that, together, these units match the load and the frequency returns to the scheduled value. However, in the process of this change the loadings of various generating units change in a manner independent of economic loading considerations. In fact, some units in the process may even get overload. Some control over loading of individual units can be exercised by adjusting the gain factors included in the signal representing integral of the area control error as fed to individual units. However, this is not satisfactory.

A satisfactory solution is achieved by using the coordination of the economic dispatch (ED) with the load frequency control (LFC). Figure 2.27 shows *an automatic generation control strategy* of ED with LFC for achieving both objectives in a coordinated manner, respectively.

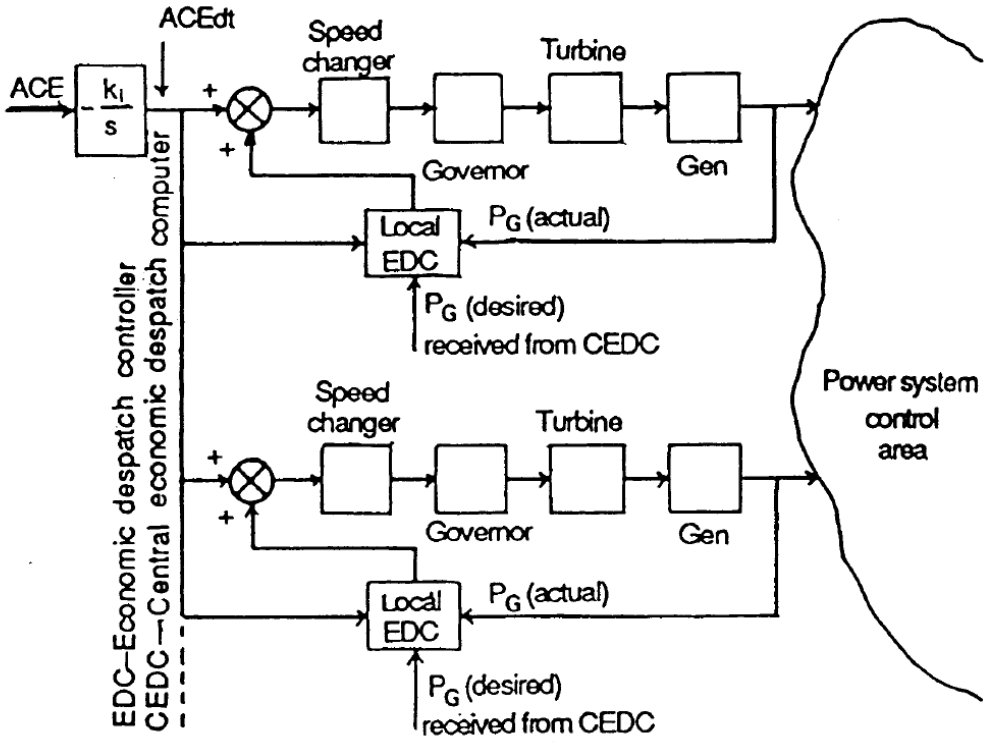


Figure 2.27 An automatic generation control strategy for coordination of ED with LFC

While the LFC is a fast acting control (a few seconds), and regulates the system around an operating point, the economic dispatch controller is a slow acting control, which adjusts the speed changer setting every minute (or half a minute) in accordance with a command signal generated by the central economic dispatch computer. Figure 2.27 shows the signal to change the speed changer setting in constructed in accordance with ED error, (P_G(desired) – P_G(actual)), suitably modified by the signal representing integral ACE (area control error) at that instant of time. The signal P_G(desired) is computed by the central economic dispatch computer (CEDC) and is transmitted to the local economic dispatch computer (EDC) installed at each power station. The system thus operates with economic dispatch error only for every short periods of time before it is readjusted.